

Forward KL Can Be Better: Normalizing Flow Boltzmann Sampling with Log-Ratio Variation

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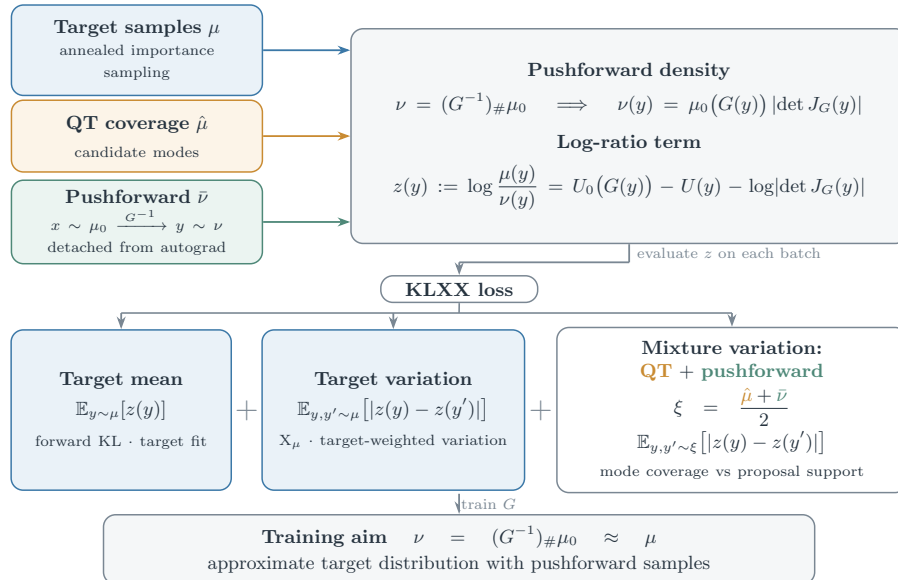
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Abstract

Normalizing flow Boltzmann generators retain a tractable proposal density, but training with forward KL depends on target samples that may be biased or omit modes. A flow can miss target mass while its observed importance weights give a high effective sample size. We introduce the log-ratio variation X_ω , the mean absolute pairwise difference of the target-to-proposal log-density ratio under a weighting measure ω , and use it to define KLXX, a new loss function for training normalizing flows: its two X's are log-ratio variations added to the forward KL, one weighted by the target surrogate, the other by a mixture of quench and temper samples with pushforward samples. Its three weighting measures thereby address target fitting, candidate mode coverage, and proposal support without changing the exact target minimizer. We derive an exponential Fisher–Rao bound and a fixed-surrogate error bound for forward KL with the target-weighted variation X_μ , then place KLXX in an adaptive-staging Boltzmann generator with exact importance reweighting and sharpening of the regularized molecular potential. We bound that generator's sampling error, so it is asymptotically unbiased. [From that bound we read](#) a propagation factor, computed from the per-stage effective sample size, that summarizes how error accumulates across stages. In the reported model and molecular tests, KLXX improves [mode-coverage ability](#) and stagewise diagnostics, and selected observables agree with independent molecular simulations. [The log-ratio variations thus supply essential information that a straightforward KL loss omits.](#)

KLXX: one log-ratio (KL), two log-ratio variations (XX)



1 Introduction

Drawing samples from a target distribution $\mu \propto \exp(-U)$ on $\mathbb{R}^d$ is a recurring problem in molecular simulation, Bayesian inference, and statistical physics (Frenkel and Smit, 2002; Robert and Casella, 2004; Landau and Binder, 2014). When μ is high-dimensional and multimodal, its mass splits among basins of U separated by barriers. Local sampling kernels, including Metropolis–Hastings (Metropolis et al., 1953; Hastings, 1970), Langevin dynamics (Langevin, 1908; Leimkuhler and Matthews, 2015), Nosé–Hoover dynamics (Nosé, 1984; Hoover, 1985), and Hamiltonian Monte Carlo (Duane et al., 1987), evolve from the current configuration and can therefore remain confined to one basin over accessible simulation times. When this occurs, finite-time estimates are biased toward the visited basin and give no direct account of the mass that was never reached.

Tempering and particle methods address this problem by connecting a tractable distribution to μ through intermediate distributions. Parallel tempering (PT) and replica exchange run copies of the system at several temperatures and exchange neighboring configurations, so a basin reached at high temperature can enter the low-temperature ensemble (Swendsen and Wang, 1986; Geyer, 1991; Hukushima and Nemoto, 1996). Annealed importance sampling (AIS) moves along a bridge of intermediate distributions while accumulating importance weights (Neal, 2001). Sequential Monte Carlo (SMC) uses a similar bridge but resamples and rejuvenates a weighted particle set at each level (Doucet et al., 2001; Del Moral et al., 2006). For a prescribed chemical process, umbrella sampling biases selected reaction coordinates to cross the relevant barriers and reweights the biased samples to recover a free energy profile (Torrie and Valleau, 1977; Kästner, 2011). Nested sampling instead rewrites the normalizing integral as a one-dimensional quadrature over a likelihood threshold (Skilling, 2006).

Recent work improves PT through three distinct mechanisms. Nonreversible swap schedules and tuned temperature grids accelerate round trips (Syed et al., 2022); variational-reference methods adapt the reference endpoint and its connecting path (Surjanovic et al., 2022); and neural transports improve communication between adjacent distributions with limited overlap (Zhang et al., 2026). Replica exchange has also been scaled beyond molecular simulation to stochastic-gradient samplers: replica exchange Langevin diffusion accelerates nonconvex optimization (Chen et al., 2019; Dong and Tong, 2021), its stochastic-gradient version corrects the swap test for noisy energy estimates (Deng et al., 2020), and later variants reduce estimator variance to restore swap rates (Deng et al., 2021). These accelerations strengthen communication between neighboring distributions and can transmit a discovered basin efficiently, but two limits remain: a basin that no chain has visited cannot enter the target ensemble (Woodard et al., 2009; Chehab et al., 2025), and the mass of a visited basin is still estimated through occupation frequencies rather than reweighting. A complementary strategy is to learn a global transport from an easily sampled reference to the target, thereby generating independent, nonlocal proposals.

Boltzmann generators implement this strategy with a learned transport map (Noé et al., 2019). Let $\mu_0 \propto \exp(-U_0)$ be an easily sampled reference distribution, usually Gaussian, and let $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be an invertible normalizing flow (Rezende and Mohamed, 2015; Papamakarios et al., 2017; Kingma and Dhariwal, 2018). We use the target-to-source convention $G_{\#}\mu \approx \mu_0$. The inverse map then sends independent source draws toward the target, and their pushforward distribution is $\nu := (G^{-1})_{\#}\mu_0$. Its density is

$$\nu(y) = \mu_0(G(y)) |\det J_G(y)|, \quad (1)$$

where $J_G(y)$ is the Jacobian matrix of G at y . Importance weights proportional to μ/ν are therefore available on pushforward samples without fitting another density or tracing a stochastic path. These weights enable importance-reweighted estimates and define the effective sample size (ESS), a diagnostic of density agreement on the support reached by ν (Liu, 2001; Doucet et al., 2001; Elvira et al., 2022). The remaining challenge is that a high ESS certifies agreement only on the reached support: a target mode absent from the pushforward samples contributes no observed weight.

The choice of loss determines how this challenge enters flow training. The original Boltzmann generator (Noé et al., 2019) and its variational-inference predecessor (Rezende and Mohamed, 2015) minimize the reverse Kullback–Leibler (KL) divergence

$$(\text{reverse}) \quad \text{KL}(\nu \parallel \mu) = \int_{\mathbb{R}^d} \nu(x) \log \frac{\nu(x)}{\mu(x)} dx, \quad (2)$$

which needs samples only from ν and evaluations of U . Its empirical gradient is therefore accessible, but the

divergence is mode-seeking (Wainwright and Jordan, 2008): a mode absent from ν contributes no samples and hence no gradient. A flow can then collapse onto a subset of the target modes (Wu et al., 2020; Felardos et al., 2023). Existing approaches interleave Markov chain Monte Carlo (MCMC) kernels with deterministic layers, encode target symmetries in the architecture, reduce gradient variance near the optimum, or anneal from a smoother potential (Wu et al., 2020; Köhler et al., 2020; Vaitl et al., 2022; Schopmans and Friederich, 2025). These changes can improve exploration, but their success still depends on whether the resulting sampler reaches each relevant mode.

The opposing loss minimizes the forward KL,

$$(\text{forward}) \quad \text{KL}(\mu \parallel \nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx, \quad (3)$$

which is mass-covering for exact distributions: it diverges if ν vanishes where μ has positive mass (Wainwright and Jordan, 2008). Its empirical form, however, needs samples from the target distribution μ being learned. Existing work resolves this circularity in several ways. Maximum likelihood uses configurations from a reference thermodynamic state (Wirnsberger et al., 2020); adaptive flow samplers train on a running Markov chain and use the flow as a global proposal (Gabri   et al., 2022); molecular flows can fit a molecular-dynamics trajectory (Klein et al., 2023); and transferable architectures amortize coverage across related molecules (Klein and No  , 2024). Persistent chains and coupled-target regression provide other approximations to the missing target expectation (Naesseth et al., 2020; Rehman et al., 2026). Because an empirical loss reflects only the regions represented by its sample source, the mass-covering property of the exact divergence does not prevent empirical mode omission. This tension motivates a loss that retains forward KL while using samples from more than one distribution.

The sampled forward KL leaves three distinct gaps. First, a mode absent from its target surrogate contributes no training points. If the flow also misses that mode, the observed weights can be nearly uniform on the reached support, so the ESS can be close to one even though target mass is absent; we call this a *fake ESS* (No   et al., 2019; Midgley et al., 2023). Second, a finite or biased target surrogate distorts the sampled loss, and the fitted flow can inherit that error. Third, target samples give little information where the proposal places excess mass, allowing leakage between modes or into target-poor regions. Addressing one of these failures does not by itself resolve the other two.

To address these gaps, we propose KLXX, a forward KL loss augmented by log-ratio variations; the graphical abstract summarizes it, and Equation (8) gives its full form. KLXX preserves the tractable target-to-proposal density ratio and exact target minimizer while separating target fitting, candidate mode coverage, and proposal support control. The first two terms of (8), the sampled forward KL and the target-weighted variation X_μ , both draw on the target surrogate supplied by annealing. The new third term mixes samples from quench and temper (QT) with pushforward samples. **QT melts the current samples, drives them into local minima of U , and re-spreads them under the target (Section 3.3).** QT contributes candidate modes missing from the surrogate, while pushforward samples expose proposal support and leakage. QT guides coverage rather than estimating physical mode weights or guaranteeing complete discovery.

The closest related method is Flow Annealed Importance Sampling Bootstrap (FAB), which couples AIS and flow training but replaces the forward KL by the α -divergence ($\alpha = 2$) (Midgley et al., 2023). Its integrand is proportional to μ^2/ν , and its exact minimizer also minimizes importance-weight variance. FAB targets μ^2/ν with AIS and reuses those samples through a prioritized replay buffer, which must be maintained across training. Related methods insert flows into SMC, anneal toward low-temperature targets, add an inference-time SMC correction, or connect disconnected regions (Arbel et al., 2021; Matthews et al., 2022; Schopmans and Friederich, 2025; Tan et al., 2025; von Klitzing et al., 2026). KLXX instead retains the forward KL and adds nonnegative log-ratio variations. Its additional cost is the QT construction and the variation evaluations, including the extra mixture batch; it uses no replay buffer. FAB’s 2-divergence gives a stronger tail penalty, whereas KLXX obtains coverage information from QT. Section 3.2 compares their sampling mechanisms.

Centered log-ratio penalties also arise in log-variance losses for learned diffusion samplers (Richter and Berner, 2024) and in log-dispersion regularization (LDR) losses for Boltzmann generators (Schopmans et al., 2026). Although the present work was developed independently of LDR, these connections place our construction within a broader family of log-ratio regularizers. Our target-weighted term differs by pairing the

forward KL integrand at two independent target points, which yields the double-integral variation in (4). KLXX further evaluates this variation under a mixture of QT and pushforward samples.

A separate line trains a stochastic process rather than a bijection. Controlled stochastic differential equations, denoising diffusions, and flow-matching models learn a drift or velocity that transports the reference toward μ (Zhang and Chen, 2022; Vargas et al., 2023; Berner et al., 2024; Nüsken and Richter, 2021; Richter and Berner, 2024). Energy-only variants such as iterated denoising energy matching (iDEM) alternate a sampling loop with a score- or velocity-fitting loop and a replay buffer (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024). Learned nonequilibrium transport augments AIS with a trainable drift, and later work extends this approach to larger many-body systems (Albergo and Vanden-Eijnden, 2025; Havens et al., 2025; Akhound-Sadegh et al., 2025).

These stochastic transports can be more expressive than the invertible architectures used here. Their terminal density, however, often requires per-sample ordinary differential equation (ODE) integration (He et al., 2025). We retain a bijection because the exact proposal density, importance ratio, ESS, and reweighting are central to the loss and final estimator. Table 1 summarizes these distinctions.

method	loss	training mechanism
reverse KL	$\text{KL}(\nu\ \mu)$	pushforward samples; no target sampler
forward KL	$\text{KL}(\mu\ \nu)$	annealed target surrogate in this work
FAB	2-divergence (μ^2/ν)	AIS on μ^2/ν ; MCMC differentiates ν ; replay buffer
LDR	$\text{KL}(\mu\ \nu) + \mathbb{E}_\mu[z - \mathbb{E}_\mu z ^p]$	fixed energy-labeled data; no extra pushforward samples
iDEM	energy matching	iterative sampler; stochastic transport; replay buffer
KLXX	$\text{KL}(\mu\ \nu) + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$	target annealing, QT, and detached pushforward samples no MCMC derivative of ν or replay buffer

Table 1: Losses and training mechanisms for the methods discussed above. Here $z = \log(\mu/\nu)$; the LDR row gives its target-weighted form with $p = 1$ for LDR-L1 and $p = 2$ for LDR-L2. The forward KL row uses the annealed baseline in this paper but may instead use external target data. The table compares how samples enter each loss; it does not assume equal computational budgets.

We now turn from these method-level comparisons to the mathematical definition of KLXX. Define the tractable log-density ratio $z(x) := \log \frac{\mu(x)}{\nu(x)}$. For a nonnegative density ω on $\mathbb{R}^d$, define the *log-ratio variation*

$$X_\omega(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x) \omega(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (4)$$

which measures how far z is from a constant on the region weighted by ω . For $\omega = \mu$, X_μ averages $|z(x) - z(y)|$ over independent target points, whereas LDR-L1 uses the centered penalty $\mathbb{E}_\mu|z - \mathbb{E}_\mu z|$. Jensen’s inequality and the triangle inequality give $\mathbb{E}_\mu|z - \mathbb{E}_\mu z| \leq X_\mu(\mu\|\nu) \leq 2\mathbb{E}_\mu|z - \mathbb{E}_\mu z|$. Thus, at the same coefficient, $\text{KL} + X_\mu$ and LDR-L1 are comparable up to a factor of two. This suggests similar training performance, although the losses are not equivalent. Since X_ω is nonnegative and vanishes at $\nu = \mu$, adding it to the exact forward KL preserves the target as a minimizer.

The target distribution μ supplies samples for both the forward KL and X_μ . The wide-coverage measure $\hat{\mu}$ supplies candidate-mode samples, while $\bar{\nu}$ is exactly the pushforward distribution ν of the current proposal, detached from the differentiation graph. In other words, $\bar{\nu}$ is a *stop-gradient* copy of ν : it supplies the same samples, but they are held fixed when the loss is differentiated. The combined loss is

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) + X_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu), \quad (5)$$

where $\lambda, \alpha, \beta \geq 0$. For simplicity, we assume $\alpha + \beta = 1$ in this section; Section 3.1 allows arbitrary nonnegative mixture weights. Under this convention, the final term is the log-ratio variation under the mixture distribution $\alpha\hat{\mu} + \beta\bar{\nu}$. Its pairwise expectation includes pairs drawn within either distribution and pairs with one sample from each distribution.

To rewrite the loss (5) in a tractable map-level form, use $\nu = (G^{-1})_\# \mu_0$, which implies the exact identity $G_\# \nu = \mu_0$. Combining (1) with $\mu(y) = \exp(-U(y))$ and $\mu_0(x) = \exp(-U_0(x))$ gives the following expression

for the log-density ratio defined above:

$$z(y) = \log \frac{\mu(y)}{\nu(y)} = U_0(G(y)) - U(y) - \log |\det J_G(y)|. \quad (6)$$

The normalizers of μ and μ_0 are absorbed into the potentials here, so (6) defines z up to an additive constant, which depends on neither y nor G . Substituting (6) into (5) yields

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu} [z(y)]}_{\text{KL}(\mu \parallel \nu)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu} [|z(y) - z(y')|]}_{X_\mu(\mu \parallel \nu)} + \underbrace{\mathbb{E}_{y, y' \sim \alpha \hat{\mu} + \beta \bar{\nu}} [|z(y) - z(y')|]}_{X_{\alpha \hat{\mu} + \beta \bar{\nu}}(\mu \parallel \nu)}. \quad (7)$$

KLXX is formally defined by the loss (7) with the default choice $(\lambda, \alpha, \beta) = (1, \frac{1}{2}, \frac{1}{2})$, and explicitly reads

$$\text{KLXX}[G] = \underbrace{\mathbb{E}_{y \sim \mu} [z(y)]}_{\text{KL}(\mu \parallel \nu)} + \underbrace{\mathbb{E}_{y, y' \sim \mu} [|z(y) - z(y')|]}_{X_\mu(\mu \parallel \nu)} + \underbrace{\mathbb{E}_{y, y' \sim (\hat{\mu} + \bar{\nu})/2} [|z(y) - z(y')|]}_{X_{(\hat{\mu} + \bar{\nu})/2}(\mu \parallel \nu)}. \quad (8)$$

With equal target and mixture batch sizes, KLXX evaluates about twice as many samples per step as forward KL and also requires the QT construction.

Our contributions are as follows. First, we introduce [a new KLXX loss, whose components mix](#) QT and pushforward samples. Pairwise differences remove the unknown target normalizer, and the exact target loss retains $\nu = \mu$ as a minimizer. The target-weighted variation reuses the forward KL batch, while the mixture supplies candidate-mode and proposal support information absent from that batch.

Second, we analyze forward KL with X_μ in Fisher–Rao geometry. The regularizer contributes a nonpositive term to the instantaneous KL dissipation in (12). For a fixed biased target surrogate and a regularization strength below the threshold of Theorem 1, it also subtracts a nonnegative discrepancy term from the [stationary error](#) bound. These results do not cover the full KLXX loss, the changing empirical surrogate used in training, or discrete neural-network optimization.

Third, we demonstrate that QT can reach target basins outside the regions covered by the source and annealed samples. In the Himmelblau and Sparse tests, the two losses trained without QT miss remote wells, while both QT-based losses recover every mode; mixing QT and pushforward samples also reduces visible intermodal leakage (Figures 3 and 4). We therefore report ESS together with coverage, since uniform weights on reached modes do not imply complete coverage.

Fourth, we build an adaptive-staging Boltzmann generator based on KLXX without any external data or replay buffer, using only oracle access to the target potential $U(x)$ and its gradient $\nabla U(x)$. Annealing and QT generate the required samples on the fly, and the MCMC transitions do not differentiate the flow density. For molecular potentials, an additional continuation sharpens a regularized interaction core toward the desired endpoint. We also bound the error of the sample set this construction returns: Theorem 2 controls the error at rate $N^{-1/2}$ for every bounded observable, with a constant independent of N that depends on the stage maps only through their weight suprema, so the scheme is asymptotically unbiased and its error is removed by enlarging the sample alone. Reading that constant stagewise defines the propagation factors, which quantify how the error accumulates across the accepted stages; the experiments report the computable factor $\hat{F}_\Sigma$, assembled from the ESS of each accepted stage.

[In addition, we develop general Python packages](#) `jflows` and `jflows-md` for normalizing flow Boltzmann sampling with molecular potentials. With a JAX backend and OpenXLA compilation, the reported 22-atom alanine dipeptide run completes all ten accepted training stages in 72.7 minutes of total persisted stage time on one NVIDIA GeForce RTX 5090, without an external target dataset or a prescribed conformational landscape.

[The rest of the paper is organized as follows:](#) Section 2 analyzes convergence and fixed-surrogate error for $\text{KL} + \lambda X_\mu$ in Fisher–Rao geometry; Appendix A gives the proofs. Section 3 defines the full KLXX construction. Section 4 proposes the adaptive-staging Boltzmann generator and bounds the error of the sample set it returns, from which the propagation factors are read; Appendix B gives the proofs. Sections 5 and 6 report the model and molecular results, respectively, and Section 7 concludes.

2 Fisher–Rao gradient flow analysis

This section analyzes $\text{KL} + \lambda X_\mu$ as a density-space flow, for simplicity, ignoring the mixture term of the QT measure $\hat{\mu}$ and the pushforward distribution $\bar{\nu}$. Starting from the full loss (5), we therefore consider

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu), \quad (9)$$

with fixed $\lambda > 0$. The only regularizer here is X_μ , and the nonlocal term of the flow is its contribution alone.

The metric determines the resulting density dynamics. In the Wasserstein-2 geometry the forward KL is not displacement convex, so the Otto–Villani (Otto and Villani, 2000) and Bakry–Émery (Bakry et al., 2014) arguments used for the reverse KL do not apply. We instead work in the Fisher–Rao (FR) geometry. We set $\text{sgn}(0) = 0$ and assume throughout that the densities are positive and sufficiently regular for the gradient calculations. The FR gradient of a smooth functional $\mathcal{F}$ and the gradient flow for $\mathcal{L}$ are

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left(\frac{\delta \mathcal{F}}{\delta \nu} - \mathbb{E}_\nu \left[\frac{\delta \mathcal{F}}{\delta \nu} \right] \right), \quad \partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{L}(\nu_t). \quad (10)$$

For the loss (9), the FR gradient flow has the explicit form

$$\partial_t \nu_t(x) = \mu(x) - \nu_t(x) + 2\lambda \mu(x) S_t(x), \quad \text{where } S_t(x) := \mathbb{E}_{y \sim \mu} \left[\text{sgn} \left(\frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right) \right]. \quad (11)$$

The local term $\mu(x) - \nu_t(x)$ is the contribution of forward KL to the FR gradient flow, while S_t is the nonlocal correlation: the update at x depends on how $\mu(x)/\nu_t(x)$ compares with the full distribution of importance weights under μ . Since $\mathbb{E}_\mu[S_t] = 0$, this term preserves total mass. Appendix A.1 derives (11).

Differentiating $\text{KL}(\mu\|\nu_t)$ along this flow, and reducing the X_μ term by a correlation identity, gives the dissipation of the forward KL:

$$\frac{d}{dt} \text{KL}(\mu\|\nu_t) = -\chi^2(\mu\|\nu_t) - \lambda \mathbb{E}_{x,y \sim \mu} \left[\left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right], \quad (12)$$

where $\chi^2(\mu\|\nu) := \mathbb{E}_\nu[(\mu/\nu - 1)^2]$ is the chi-squared divergence. The regularizer contributes a nonpositive term to the instantaneous KL dissipation. Since $\chi^2(\mu\|\nu_t) \geq \text{KL}(\mu\|\nu_t)$, Grönwall’s lemma yields

$$\text{KL}(\mu\|\nu_t) \leq e^{-t} \text{KL}(\mu\|\nu_0), \quad t \geq 0, \quad (13)$$

without assuming log-concavity, a spectral gap, or a growth condition on μ . This is a statement in density space, and it does not imply that $\text{KL} + \lambda X_\mu$ trains faster than forward KL. The additional term in (12) describes only how X_μ changes the density flow, rather than a universal transient speedup.

The convergence result assumes that the outer expectation is taken under the exact target. However, the batch of target samples generated by AIS is usually biased (see Algorithm 1). To isolate surrogate bias, fix a smooth density $\tilde{\mu}$ and replace μ by $\tilde{\mu}$ only in the outer integrals while retaining the exact pointwise log-ratio. This defines

$$\tilde{\mathcal{L}}[\nu] = \mathbb{E}_{x \sim \tilde{\mu}} \left[\log \frac{\mu(x)}{\nu(x)} \right] + \lambda \mathbb{E}_{x,y \sim \tilde{\mu}} \left[\left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| \right]. \quad (14)$$

We assume that the densities are positive on their common domain and that the displayed KL, chi-squared, and pair integrals are finite. At $\lambda = 0$, the stationary density is $\nu^* = \tilde{\mu}$, so the residual error is $\text{KL}(\mu\|\tilde{\mu})$. The next result shows when X_μ improves this stationary error bound.

Theorem 1 *Assume $\lambda \in [0, \frac{1}{2})$. Then, under the positivity and finiteness conditions above, every stationary density ν^* of the gradient flow for the fixed-surrogate loss $\tilde{\mathcal{L}}$ in (14) satisfies*

$$\text{KL}(\mu\|\nu^*) \leq \text{KL}(\mu\|\tilde{\mu}) - \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy. \quad (15)$$

Within the theorem’s assumptions, the correlation induced by the second term in (14) raises ν^* in comparatively underweighted regions and lowers it elsewhere. Here $\lambda < \frac{1}{2}$ is a sufficient condition for the theorem, and the experiments take the more aggressive choice $\lambda = 1$. Appendix A gives the variation, correlation identity, and proofs.

3 Flow training with log-ratio variation

This section turns the abstract loss of the introduction into a training algorithm using a flow map G . Let U_0 and U be the source and target potentials on $\mathbb{R}^d$, so that $\mu_0 \propto \exp(-U_0)$ is an easy-to-sample reference and $\mu \propto \exp(-U)$ is the target. We parameterize $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ so that $G_{\#}\mu \approx \mu_0$ and write $\nu := (G^{-1})_{\#}\mu_0$ for the induced approximation to μ . The loss itself evaluates only the native target-to-source map G and its log-Jacobian; it does not explicitly involve G^{-1} . The inverse map G^{-1} nevertheless generates pushforward samples used both to initialize AIS and to evaluate the $\bar{\nu}$ term; these G^{-1} evaluations are sampling-only. The sampled coordinates are detached from the inverse-map path, but the log-ratio evaluations at those coordinates are still differentiated through the native map G . We first assemble the training loss and identify its sampling ingredients (Section 3.1), then supply them in turn: an AIS approximation of μ (Section 3.2) and the quench and temper construction of the wide-coverage measure $\hat{\mu}$ (Section 3.3). Section 3.4 gives the ESS and coverage diagnostics used to evaluate the trained flow.

3.1 Training loss with log-ratio variation

By change of variables, the pushforward density is $\nu(y) = \mu_0(G(y))|\det J_G(y)|$, so every term is a functional of the scalar field $z(y)$ in (6):

$$z(y) = \log \frac{\mu(y)}{\nu(y)} = U_0(G(y)) - U(y) - \log |\det J_G(y)|.$$

The forward KL is its mean under the target, $\text{KL}(\mu\|\nu) = \mathbb{E}_{y \sim \mu}[z(y)]$. Set $\vartheta := \alpha + \beta$. For $\vartheta > 0$, let $\xi := (\alpha\hat{\mu} + \beta\bar{\nu})/\vartheta$ be the normalized mixture, with any zero-weight component omitted. Since $\alpha\hat{\mu} + \beta\bar{\nu} = \vartheta\xi$, the double integral in (4) produces the factor ϑ^2 below. The map-level loss is

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu}[z(y)]}_{\text{KL}(\mu\|\nu)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu}[|z(y) - z(y')|]}_{X_{\mu}(\mu\|\nu)} + \underbrace{\vartheta^2 \mathbb{E}_{y, y' \sim \xi}[|z(y) - z(y')|]}_{X_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu)}. \quad (16)$$

When $\vartheta = 0$, the last term is omitted. The default choice $\lambda = 1$ and $\alpha = \beta = \frac{1}{2}$ defines KLXX. The pushforward distribution $\nu = (G^{-1})_{\#}\mu_0$ depends on the trainable parameters of G . The barred measure $\bar{\nu}$ is the same distribution frozen at the current parameters: samples are drawn through G^{-1} and detached from that sampling path, but z at the sampled coordinates is still differentiated through the native map G .

The three sampling measures entering (16) address complementary failure modes of forward KL:

- μ : *target accuracy*. The term X_{μ} penalizes pairwise log-ratio variation within the target batch. Theorem 1 gives its density-space bound for $\lambda < \frac{1}{2}$; the finite-training effect at $\lambda = 1$ is tested empirically.
- $\hat{\mu}$: *candidate mode coverage*. QT provides log-ratio comparisons on basins represented in its wide-coverage set, including basins absent from the annealed batch. The melt scale m_e in Algorithm 2 controls how broadly QT searches.
- $\bar{\nu}$: *proposal support*. Pushforward samples evaluate the log-ratio where the current proposal places mass. The term is intended to reduce variation in target-poor regions; the experiments test whether this is accompanied by less visible leakage.

Adding these nonnegative terms to the exact target forward KL preserves $\nu = \mu$ as a minimizer. Both $z(y)$ and $\bar{\nu}$ depend explicitly on the flow G , but only the dependence through $z(y)$ enters the differentiation graph of the loss $\mathcal{L}[G]$ during training.

The forward KL and X_{μ} share one target batch. Evaluate $\mathbf{z} = (z(y_1), \dots, z(y_B))$, use $\text{mean}(\mathbf{z})$ for the forward KL, and use $\text{mean}(|\mathbf{z} - \mathbf{z}[\sigma]|)$ for X_{μ} , where σ is a uniformly random batch permutation. This estimator introduces no additional flow evaluation or differentiation because it reuses the same $\mathbf{z}$ and the same flow differentiation graph; the only added operations are the random permutation and pairwise difference. When $\vartheta > 0$, the last term evaluates one additional batch drawn from ξ , applies the same permuted-pair estimator, and multiplies it by ϑ^2 . At the KLXX setting, each sample in this additional batch is drawn from $\hat{\mu}$ or $\bar{\nu}$ with probability $1/2$.

The loss draws on three measures, but $\bar{\nu}$ requires no construction beyond source draws, so at most two must be built. The $\bar{\nu}$ batch consists of pushforward samples $\bar{y}_i = G^{-1}(x_i)$ generated from source samples $x_i \sim \mu_0$. The inverse-map sampling path is detached, while $z(\bar{y}_i)$ is then evaluated through the native map G and differentiated normally. An approximation of the target μ for the forward KL and X_μ terms is supplied by the AIS construction in Section 3.2. When $\alpha > 0$, a second construction supplies samples of the wide-coverage measure $\hat{\mu}$ by quench and temper (Section 3.3). Throughout loss differentiation, the sampled coordinates and all resampling, mixture, and permutation choices are treated as fixed; gradients pass only through evaluations of z at the resulting coordinates. With equal target and mixture batch sizes, $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ evaluates about twice as many samples per optimizer iteration as forward KL. This does not imply twice the total training time: constructing $\hat{\mu}$ also applies quench and temper to the validation set and adds a problem- and stage-dependent cost. Inference uses the same flow architecture for both methods.

Two recurring losses are $\text{KL} + X_\mu$, with $\lambda = 1$ and $\alpha = \beta = 0$, and KLXX , with $\lambda = 1$ and $\alpha = \beta = \frac{1}{2}$. The first adds no sampling measure beyond the target batch, while the second draws the two components of its additional batch with equal probability. We report the full KLXX loss as $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ when it must be distinguished from the other ablations.

Remark 1 *We do not train with log-ratio variation alone, because empirical pairwise terms constrain relative log-ratios only on sampled regions and provide no global density-shaping constraint elsewhere. Appendix D.2 of the LDR study (Schopmans et al., 2026) shows the analogous failure: training with LDR-L1 or LDR-L2 alone, without a KL term, produces qualitatively incorrect distributions. Accordingly, every X-regularized loss in this paper includes forward KL.*

3.2 Approximating μ by annealed importance sampling

Evaluating the μ -expectations in (16) requires samples from μ , which are not directly available. We form a surrogate by drawing pushforward samples $y_1, \dots, y_B$ from $\nu = (G^{-1})_{\#}\mu_0$ and reweighting them. The importance weight is

$$w(y) = \exp\left(U_0(G(y)) - U(y) - \log|\det J_G(y)|\right) = \frac{\mu(y)}{\nu(y)}, \quad y \in \mathbb{R}^d. \quad (17)$$

In high dimensions, however, μ and ν can differ greatly, so the weights can collapse onto a few outliers. We temper the reweighting through $M + 1$ bridge distributions

$$\pi_m(y) \propto \nu(y)^{1-\frac{m}{M}} \mu(y)^{\frac{m}{M}} \propto ((G^{-1})_{\#}\mu_0)(y) w(y)^{\frac{m}{M}}, \quad m = 0, 1, \dots, M, \quad (18)$$

so that $\pi_0 = \nu$ and $\pi_M = \mu$. Each level reweights by only $w(y)^{1/M}$, reducing the degeneracy of direct importance sampling. After each reweighting, we resample and diffuse the duplicates with the Metropolis-adjusted Langevin algorithm (MALA) (Roberts and Tweedie, 1996) before the next level.

Algorithm 1: Annealed importance sampling (AIS) for a biased target surrogate

Input: pushforward samples $y_1, \dots, y_B$ from $(G^{-1})_{\#}\mu_0$, target distribution μ , ladder length M

Output: biased target-surrogate batch $y_1, \dots, y_B$ approximating μ

```

1 for  $m = 1, \dots, M$  do
2   Compute the incremental weights  $w(y_i)^{1/M} = \exp(z(y_i)/M)$  on  $y_1, \dots, y_B$ ;
   // resample
3   Resample from the weighted set,  $y'_1, \dots, y'_B \leftarrow \text{resample}(y_{1:B}, (w(y_i)^{1/M})_{i=1}^B)$ ;
   // rejuvenate under the target
4   Diffuse the resampled duplicates by a short MALA chain on  $\mu$ ,  $y_{1:B} \leftarrow \text{MALA}(y'_{1:B}, U)$ ;
5 end
```

A standard SMC sampler targeting the path (18) would rejuvenate at level m with a Markov kernel invariant

for π_m . For $m < M$, a MALA kernel for π_m would require $\nabla_y \log \nu(y)$ and hence positional derivatives of the flow density, including $\nabla_y \log |\det J_G(y)|$; at $m = M$, this term disappears because $\pi_M = \mu$. Algorithm 1 instead applies MALA directly under μ , using only $-\nabla U$ in its proposal. MALA preserves μ , but not the intermediate π_m , so the output is a biased target surrogate. The construction trades intermediate invariance for a sampler that moves resampled samples under the final target.

The AIS construction in Algorithm 1 differs from FAB (Midgley et al., 2023). FAB places the flow inside the chains, and its MCMC transitions target the surrogate μ^2/ν , whose log-density gradient is

$$\nabla_y \log \frac{\mu^2(y)}{\nu(y)} = -2\nabla U(y) - \nabla_y \log \nu(y),$$

and therefore contains the positional gradient of the pushforward density at every MCMC step. FAB trains from a prioritized replay buffer of weighted AIS samples with a staleness correction. Reusing this buffer reduces repeated AIS generation but adds storage, prioritized sampling, and correction work. Here every MALA chain instead targets μ and uses only $-\nabla U$; the flow enters through the initial pushforward samples and evaluations of z . Each gradient step starts from a new source batch, so no replay buffer is kept.

The exploration costs also differ. Relative to forward KL, KLXX adds the QT construction and pairwise log-ratio variation evaluations: X_μ reuses the target batch, while the mixture term evaluates an additional batch. The fixed N -sample QT set of Section 3.3 supplies separate coverage information, whereas FAB relies on its AIS budget for exploration. FAB’s 2-divergence gives it a stronger tail-coverage incentive.

Remark 2 *In principle, AIS or another SMC sampler can propagate from the source μ_0 to the target μ on its own, with no flow at all. Two drawbacks argue against that choice. First, such a sampler explores only where its own trajectories go, so nothing guarantees coverage of the target modes; the QT set of Section 3.3 is what supplies candidate modes in KLXX. Second, with no flow to deform the proposal toward the next distribution, each level reweights and resamples at a much lower ESS, and repeated resampling of a degenerate weight set raises the Monte Carlo error of the returned samples.*

The Boltzmann generator of Section 4 compares each trained flow with the identity proposal and keeps whichever has the larger validation ESS, so an accepted stage is never worse than the identity in that diagnostic, up to the finite validation sample. A trained Boltzmann generator therefore has no reason to perform systematically worse than the identity, which propagates by reweighting, resampling, and MALA alone. Section 6.3 reports the empirical side: the KLXX Boltzmann generator has the smallest propagation factor in all ten recorded alkane settings, while forward KL approaches the identity result as the molecule grows.

3.3 Constructing $\hat{\mu}$ by quench and temper

The last term of the loss (16) uses the wide-coverage measure $\hat{\mu}$ to improve candidate mode coverage when $\alpha > 0$. A finite source batch can omit remote target basins, and annealing initialized from that batch provides no log-ratio comparisons there unless its trajectories reach them.

QT addresses this limitation through three steps. The *melt* scatters the input samples across a wider region, the *quench* drives them toward mode centers with the limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) method (Liu and Nocedal, 1989), and the *temper* spreads them under the target around the discovered modes. Together, these steps construct a wide-coverage set that can include remote basins; $X_{\hat{\mu}}$ then adds comparisons on basins absent from the annealed batch, strengthening mode discovery beyond annealing alone. Formally, QT is the map

$$\hat{\mathcal{Y}} = \text{QT}(\mathcal{Y}, U)$$

from the validation set $\mathcal{Y} = \{x_1, \dots, x_N\}$ and target potential U to the N QT samples $\hat{\mathcal{Y}} = \{\hat{x}_1, \dots, \hat{x}_N\}$. Algorithm 2 applies this map once per stage attempt. The output $\hat{\mathcal{Y}}$ is held fixed as a reservoir during training. At each training step a mini-batch of B samples is drawn uniformly from that reservoir and refreshed by a short MALA chain before entering the computation. We use $\hat{\mu}$ for the law of this draw-and-refresh operation; its input is the uniform empirical measure on the fixed QT reservoir.

On a torus or Euclidean–torus product, periodic coordinates are interpreted modulo their period after the melt and after each MALA proposal. If the QT set contains a basin absent from the annealed target batch,

Algorithm 2: Quench and temper (QT): wide-coverage construction

Input: validation set $\mathcal{Y} = \{y_1, \dots, y_N\}$, target potential U , melt scale m_e
Output: wide-coverage set $\hat{\mathcal{Y}} = \{\hat{y}_1, \dots, \hat{y}_N\}$ forming $\hat{\mu}$, written $\hat{\mathcal{Y}} = \text{QT}(\mathcal{Y}, U)$
// melt
1 Scatter the samples $y_i \leftarrow y_i + m_e \zeta_i$ with $\zeta_i \sim \mathcal{N}(0, I_d)$ i.i.d. and melt scale $m_e > 0$;
// quench
2 Drive each sample toward a mode center of μ by L-BFGS on U , updating $y_1, \dots, y_N$;
// temper
3 Spread the samples around each mode, $\hat{y}_{1:N} \leftarrow \text{MALA}(y_{1:N}, U)$;

$X_{\hat{\mu}}$ supplies pairwise log-ratio evaluations on that basin. A large melt scale m_e can bias the QT set toward basins far from the input set. This is intentional: $\hat{\mu}$ is a coverage measure, not an estimate of target mode probabilities. Adding $X_{\hat{\mu}}$ to the forward KL preserves $\nu = \mu$ as a minimizer, while empirical frequencies in the finite QT set affect the optimization path.

3.4 Evaluation: ESS, coverage, and importance sampling

We assess the quality of the pushforward samples through two complementary diagnostics. The ESS measures density agreement through the observed importance weights on support reached by the flow. Coverage instead checks whether the pushforward samples reach the regions represented by a fixed QT reference set. A high ESS can coexist with a missing mode, while high coverage does not establish the correct relative density within the reached regions. We therefore report the two diagnostics together.

At evaluation time the trained flow G supplies the pushforward distribution $\nu = (G^{-1})_{\#} \mu_0$ and its tractable density $\nu(y) = \mu_0(G(y)) |\det J_G(y)|$. The importance weight $w(y)$ in (17) can therefore be evaluated on pushforward samples without fitting another density model. Draw $\mathbf{y} = (y_1, \dots, y_N)$ with $y_i = G^{-1}(x_i)$ and $x_i \sim \mu_0$ independently, and write $\mathbf{w} = (w_1, \dots, w_N)$ for the corresponding weights $w_i = w(y_i)$.

The first diagnostic is the ESS, normalized to $(0, 1]$ by dividing the classical ESS by the sample count N . For the weight vector $\mathbf{w}$, define

$$\text{ESS}(\mathbf{w}) = \frac{(\sum_{i=1}^N w_i)^2}{N \sum_{i=1}^N w_i^2} \in (0, 1]. \quad (19)$$

The upper bound follows from Cauchy–Schwarz and is attained when all weights are equal. The factor N distinguishes this normalized ESS from the classical count $(\sum_i w_i)^2 / \sum_i w_i^2 \in [1, N]$ (Elvira et al., 2022). A high ESS indicates uniform observed weights only on support reached by ν ; a target mode absent from the sample cannot lower it.

To detect such missing modes, we pair the ESS with the coverage diagnostic of Naeem et al. (2020). Let $\hat{\mathbf{y}} = (\hat{y}_1, \dots, \hat{y}_N) \in \mathbb{R}^{N \times d}$ be the fixed QT reference set produced by Algorithm 2. For each $1 \leq i \leq N$, the κ -nearest-neighbor radius of $\hat{y}_i$ (with $\kappa = 5$) is

$$\text{NND}_{\kappa}(\hat{y}_i; \hat{\mathbf{y}}) = \inf \left\{ r \geq 0 : \#\{1 \leq j \leq N : 0 < |\hat{y}_i - \hat{y}_j| < r\} \geq \kappa \right\}, \quad (20)$$

i.e., the distance from $\hat{y}_i$ to its κ -th nearest neighbor in $\hat{\mathbf{y}}$. The coverage of $\mathbf{y}$ in $\hat{\mathbf{y}}$ is then

$$\text{Coverage}_{\kappa}(\mathbf{y}; \hat{\mathbf{y}}) = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left\{ \exists 1 \leq j \leq N : |y_j - \hat{y}_i| \leq \text{NND}_{\kappa}(\hat{y}_i; \hat{\mathbf{y}}) \right\}, \quad (21)$$

i.e., the fraction of points in $\hat{\mathbf{y}}$ whose κ -nearest-neighbor ball contains any of the pushforward samples. A coverage close to 1 indicates that the finite QT reference set is well covered by the flow, while a coverage well below 1 exposes missing regions that the ESS alone cannot see. Because the reference set is itself produced by QT, the diagnostic checks only modes found by that reference construction.

After these diagnostics assess the proposal, the same weights used for ESS define the resampling step. Normalize them as $p_i = w_i / \sum_{j=1}^N w_j$, resample N times from the pushforward samples with probabilities

p_i , and apply a short MALA chain under μ to rejuvenate the resulting duplicates. The resampled samples form an unweighted approximation of the target-weighted empirical measure, while MALA produces fresh configurations and retains μ as its invariant distribution.

Under the support condition $\mu \ll \nu$ and standard consistency conditions, the target-weighted empirical measure and its resampled particle approximation converge to μ as $N \rightarrow \infty$ (Liu, 2001; Doucet et al., 2001). The μ -invariant MALA refresh preserves this target limit.

4 Adaptive-staging Boltzmann generator

A single flow can be inadequate when the source-to-target change is too large, so we divide the interpolation into stages. Each stage changes the potential by a controlled amount and trains one flow with the loss in (16), using user-chosen parameters $\lambda, \alpha, \beta \geq 0$. The KLXX Boltzmann generator is the special case $(\lambda, \alpha, \beta) = (1, \frac{1}{2}, \frac{1}{2})$. Sampling diagnostics determine the stage schedule.

4.1 Why staging must be adaptive

Two obstacles prevent a single flow from carrying μ_0 to μ in one map.

- *Curse of dimensionality.* In low dimension a unimodal source can often be reshaped into a multimodal target by one map, but as d grows the target landscape develops more basins, separated by higher barriers and occupying a smaller fraction of the source support. A normalizing flow of fixed depth and width can represent only a bounded change of shape, so the deformation it realizes in one training run is limited by its capacity rather than by the training budget. Requesting the whole source-to-target change from a single flow therefore leaves part of the target unfitted in exactly the high-dimensional, multimodal problems of interest. Splitting the interpolation into stages keeps each requested deformation within the capacity of one flow, and the number of stages needed to do so grows with dimension.
- *Structural limitation of a diffeomorphism.* A trained flow is a diffeomorphism of $\mathbb{R}^d$, so the pushforward ν inherits the connected support of μ_0 : the map deforms that support continuously and cannot split it into disconnected components. Mass therefore remains on the paths joining the target basins even when the basins themselves are well fitted, an effect visible between the isolated modes of the Sparse target in Figure 4. Reweighting and MALA rejuvenation at the interpolating distributions remove the leaked samples at every stage, so the support changes shape gradually across the schedule.

The stage schedule cannot be fixed in advance, because we assume no external information about the target. Training uses no target dataset, no reference trajectory, and no prior knowledge of the number, location, or depth of the potential wells. Under this restriction the difficulty of a candidate stage cannot be read off the potential before attempting it. The algorithm therefore probes the interpolation parameter: it proposes a stage increment, trains the stage flow, accepts the increment only when the resulting proposal passes a sampling gate on the validation set, and otherwise shrinks the increment and retries. Adaptive staging is thus more than a convenience. It is the mechanism that turns oracle access to $U(x)$ and $\nabla U(x)$ into a stage schedule matched to the difficulty the target actually presents, and it lets easy regions advance in few large stages while hard regions receive as many small stages as they require.

4.2 Construction of the adaptive stage schedule

To interpolate from the source $\mu_0 \propto \exp(-U_0)$ to the target $\mu \propto \exp(-U)$, we seek an increasing sequence

$$0 = t_0 < t_1 < \dots < t_K = 1$$

defining the interpolating potential and measure

$$U_t(x) = (1-t)U_0(x) + tU(x), \quad \mu_t(x) \propto \exp(-U_t(x)), \quad t \in [0, 1].$$

Here t is a dimensionless interpolation parameter. The discrete values t_k are stage points, and stage k is the transition from t_{k-1} to t_k . At the accepted stage points, write $\mu_k := \mu_{t_k}$, so μ_0 is the source and μ_K is the

target. Stage k trains a target-to-source map G_k , whose inverse induces the proposal $\nu_k = (G_k^{-1})_{\#}\mu_{k-1}$ for μ_k . The number of stages K is selected during the run. The central scheduling choice is the stage increment $\Delta t_k = t_k - t_{k-1}$: a large increment can be hard to fit, whereas a small increment adds stages.

Training G_k requires the target surrogate for μ_k defined in Section 3.2. Before training, Algorithm 3 can screen the proposed stage increment Δt_k with an optional classical SMC calculation. Its smallest per-level ESS measures overlap along that candidate stage for the current sample set. This is a difficulty diagnostic, not a mode-coverage guarantee. The screen floor τ_{SMC} lies in $[0, 1)$, and setting it to zero disables the screen. Throughout this section, the validation set is the sample set carried across stages and used in full for training batches, QT, and stage selection; it is not a held-out test set, and neither QT nor the SMC screen first draws a separately sized subset of it.

Algorithm 3: Adaptive selection of the stage increment Δt_k

Input: potentials U_0, U , stage point t_{k-1} , validation set $\mathcal{Y}_{k-1} = \{y_1, \dots, y_N\}$ approximating μ_{k-1} , ladder length M , SMC ESS floor $\tau_{\text{SMC}} = 0$, shrink factor $\gamma = 0.7$, initial guess t_{init}

Output: next stage point t_k , giving the stage increment $\Delta t_k = t_k - t_{k-1}$

```

1  $U_{t_{k-1}} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U;$ 
   // start from the supplied initial guess
2  $t_k \leftarrow t_{\text{init}};$ 
3 if  $\tau_{\text{SMC}} = 0$  then
4   | return  $t_k;$ 
5 end
6 while true do
   | // run a classical SMC ladder across the candidate stage
7   | Run an  $M$ -level SMC ladder on the set  $\mathcal{Y}_{k-1}$ , using the linear potential bridge from  $U_{t_{k-1}}$  to  $U_{t_k}$ ;
   | // evaluate the ladder by its worst per-level ESS
8   | Record  $\text{ESS}_{\min}$ , the smallest per-level ESS over the  $M$  ladder levels;
9   | if  $\text{ESS}_{\min} \geq \tau_{\text{SMC}}$  then
10  | | return  $t_k;$ 
11  | else
   | | // shrink the stage increment toward  $t_{k-1}$ 
12  | |  $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1});$ 
13  | end
14 end
```

These SMC ladders involve only the potentials U_0 and U . They use no QT samples, so their per-level ESS can overlook a mode the SMC samples never reach. We therefore use this ESS only to screen a candidate stage; it is a heuristic measure of stage difficulty, not a coverage certificate. The selected-proposal validation in step (v) below supplies the acceptance gate, while QT supplies the separate wide-coverage set when $\alpha > 0$. By default $\tau_{\text{SMC}} = 0$, so Algorithm 3 returns t_{init} without running SMC. For a positive floor, it returns the first candidate stage along the shrink sequence whose estimated ESS clears that floor. The shrink factor is $\gamma = 0.7$. As t_k approaches t_{k-1} , the stage becomes trivial and its ESS approaches 1, so a floor below 1 is attainable under repeated shrinking in exact arithmetic.

4.3 The full training procedure

Algorithm 4 combines the stage-selection rule with the training procedure. It carries across stages a fixed-size validation set $\mathcal{Y}_k$, whose empirical measure approximates the per-stage target μ_k . Each candidate stage builds its QT set when $\alpha > 0$, trains a flow with the selected loss weights, and compares that flow with the identity proposal. If neither proposal reaches the validation ESS floor $\tau_v \in [0, 1)$, the algorithm shrinks

the stage increment and retries. The displayed loop ends when $t_K = 1$. Practical implementations use an endpoint tolerance and finite caps on retries and stages; reaching a cap returns an incomplete stage schedule.

- (i) *Validation set.* Stage k uses the *validation set size* N and the *batch size* B , taken at least 2 so that the permuted-pair estimator has a pair to form. The validation set $\mathcal{Y}_{k-1} = \{y_1, \dots, y_N\} \approx \mu_{k-1}$ is carried over from the previous stage. Training batches are drawn at run time, B samples per gradient step, from $\mathcal{Y}_{k-1}$; no separate training set is built.
- (ii) *Adaptive stage selection.* Algorithm 3 selects the next stage point t_k . The first stage starts from $t_{\text{init}} = t_{\text{safe}} = 0.2$. Later stages extrapolate the last accepted stage increment, $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$, with enlarge factor $\Gamma = 1.5$. When a positive floor is supplied, the algorithm shrinks the stage increment until the classical SMC ESS on all N samples clears that floor. The selected stage target is $\mu_k \propto \exp(-U_{t_k})$, with $U_{t_k} = (1 - t_k)U_0 + t_k U$.
- (iii) *Wide-coverage set.* When $\alpha > 0$, quench and temper (Algorithm 2) acts on the validation set $\mathcal{Y}_{k-1}$ under U_{t_k} . The resulting N samples $\tilde{\mathcal{Y}}_k = \text{QT}(\mathcal{Y}_{k-1}, U_{t_k})$ form the fixed stage reservoir. The measure $\hat{\mu}_k$ is the law obtained by drawing uniformly from this reservoir and applying the short MALA refresh used at each gradient step; it supplies the wide-coverage component of the last term in (22).
- (iv) *Training loss.* Each attempt initializes G_k to the identity map and trains it by the general loss (16) specialized to the stage $\mu_{k-1} \rightarrow \mu_k$. With the pushforward distribution $\nu_k = (G_k^{-1})_{\#}\mu_{k-1}$ and its frozen sampling measure $\bar{\nu}_k$, the stage log-ratio is

$$z_k(y) = U_{t_{k-1}}(G_k(y)) - U_{t_k}(y) - \log |\det J_{G_k}(y)|.$$

Let $\vartheta = \alpha + \beta$. For $\vartheta > 0$, define the stage specialization of the normalized mixture ξ ,

$$\xi_k := \frac{\alpha \hat{\mu}_k + \beta \bar{\nu}_k}{\vartheta}.$$

The stage loss then reads

$$\mathcal{L}[G_k] = \underbrace{\mathbb{E}_{y \sim \mu_k} [z_k(y)]}_{\text{KL}(\mu_k \parallel \nu_k)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu_k} [|z_k(y) - z_k(y')|]}_{X_{\mu_k}(\mu_k \parallel \nu_k)} + \vartheta^2 \underbrace{\mathbb{E}_{y, y' \sim \xi_k} [|z_k(y) - z_k(y')|]}_{X_{\alpha \hat{\mu}_k + \beta \bar{\nu}_k}(\mu_k \parallel \nu_k)}. \quad (22)$$

The μ_k -expectations are estimated from the biased annealed surrogate: a batch $\mathcal{X}$ drawn from $\mathcal{Y}_{k-1}$ is mapped to the batch of pushforward samples $\bar{\mathcal{X}}_\nu = G_k^{-1}(\mathcal{X})$, approximating ν_k , and annealed toward μ_k by Algorithm 1 with $G = G_k$. The batch $\mathcal{X}_{\xi_k}$ is formed by drawing from $\hat{\mu}_k$ with probability α/ϑ , and from $\bar{\nu}_k$ with probability β/ϑ ; the permuted-pair estimate is then multiplied by ϑ^2 . The choices $(\lambda, \alpha, \beta) = (0, 0, 0)$ and $(1, 0, 0)$ give forward KL and KL+ X_μ , respectively; $(1, 1, 0)$ gives KL+ X_μ + $X_{\hat{\mu}}$, and $(1, \frac{1}{2}, \frac{1}{2})$ gives the default KLXX loss.

- (v) *Validation set update.* Once the candidate G_k is trained, each $y \in \mathcal{Y}_{k-1}$ is pushed through its inverse to $\tilde{y} = G_k^{-1}(y)$. This gives the trained set of pushforward samples $\tilde{\mathcal{Y}}_k^{\text{tr}} = G_k^{-1}(\mathcal{Y}_{k-1})$ from $\nu_k = (G_k^{-1})_{\#}\mu_{k-1}$, together with the importance weights

$$w_k^{\text{tr}}(\tilde{y}) := \exp(z_k(\tilde{y})), \quad \tilde{y} \in \tilde{\mathcal{Y}}_k^{\text{tr}}.$$

The identity proposal keeps $\mathcal{Y}_{k-1}$ unchanged and has weights $w_k^{\text{id}}(y) = \exp(U_{t_{k-1}}(y) - U_{t_k}(y))$. We select the pair with the larger ESS over the validation set and denote it by $(\tilde{\mathcal{Y}}_k, w_k)$, taking the stage map to be G_k or the identity accordingly. If $\text{ESS}(w_k) < \tau_v$, with default $\tau_v = 0.6$, we discard the candidate, shrink to $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$, and repeat steps (iii)–(v) at the reduced stage increment. Otherwise we advance the validation set to $\mathcal{Y}_k = \text{MALA}(\text{resample}(\tilde{\mathcal{Y}}_k, w_k; N), U_{t_k}) \approx \mu_k$.

The accepted stage schedule defines a Boltzmann generator for μ . The process starts with samples from μ_0 . At stage k , it applies the selected inverse map or identity, computes w_k , resamples, and rejuvenates under U_{t_k} . The resulting sample set supplies stage $k + 1$.

Algorithm 4: Adaptive-staging Boltzmann generator training

Input: potentials U_0, U ; sizes $N, B \geq 2$; loss weights $\lambda, \alpha, \beta \geq 0$; ladder length M ; ESS floors $\tau_{\text{SMC}} = 0$, $\tau_v = 0.6$; factors $\gamma = 0.7$, $\Gamma = 1.5$; safe start $t_{\text{safe}} = 0.2$

Output: completed stage schedule $0 = t_0 < \dots < t_K = 1$ and selected stage maps $\{G_k\}_{k=1}^K$

// validation set, $\mathcal{Y}_k \approx \mu_k$

```
1  $\mathcal{Y}_0 \leftarrow N$  samples from  $\mu_0$ ,  $t_0 \leftarrow 0$ ,  $k \leftarrow 0$ ;  
2 while  $t_k < 1$  do  
3    $k \leftarrow k + 1$ ;  
4   Set  $U_{t_{k-1}} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U$ ;  
   // (i) validation set  
5   Use  $\mathcal{Y}_{k-1}$  for stage selection, validation, and QT when  $\alpha > 0$ ;  
   // (ii) adaptive stage selection  
6   if  $k = 1$  then  
7     Set  $t_{\text{init}} = t_{\text{safe}}$ ;  
8   else  
9     Set  $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$ ;  
10  end  
11  Choose  $t_k$  by Algorithm 3 on  $\mathcal{Y}_{k-1}$  from  $t_{\text{init}}$ ;  
12  Set  $U_{t_k} \leftarrow (1 - t_k)U_0 + t_kU$ ;  
13  repeat  
14    Initialize  $G_k$  to the identity map;  
    // (iii) optional wide-coverage set  
15    If  $\alpha > 0$ , set  $\hat{\mathcal{Y}}_k \leftarrow \text{QT}(\mathcal{Y}_{k-1}, U_{t_k})$  by Algorithm 2;  
    // (iv) general log-ratio-variation loss  
16    for each gradient step do  
17      Draw  $B$  samples  $\mathcal{X}$  from  $\mathcal{Y}_{k-1}$ ;  
18      Set the batch of pushforward samples  $\bar{\mathcal{X}}_\nu \leftarrow G_k^{-1}(\mathcal{X})$  and detach its sampling path;  
19      Anneal these pushforward samples toward  $\mu_k$  to form the biased target batch  $\mathcal{X}_\mu$  (Algorithm 1);  
20      If  $\alpha > 0$ , draw  $\hat{\mathcal{X}}_\mu \subset \hat{\mathcal{Y}}_k$  and refresh it by MALA under  $U_{t_k}$ ;  
21       $\vartheta \leftarrow \alpha + \beta$ ; if  $\vartheta > 0$ , mix  $\hat{\mathcal{X}}_\mu$  (when  $\alpha > 0$ ) and  $\bar{\mathcal{X}}_\nu$  with probabilities  $\alpha/\vartheta, \beta/\vartheta$  to form  $\mathcal{X}_{\xi_k}$ ;  
22      Update  $G_k$  with the loss below using random batch permutations; omit its last term if  $\vartheta = 0$ :  
  
      
$$\hat{\mathcal{L}}[G_k] = \hat{\mathbb{E}}_{y \in \mathcal{X}_\mu} [z_k(y)] + \lambda \hat{\mathbb{E}}_{y, y' \in \mathcal{X}_\mu} [|z_k(y) - z_k(y')|] + \vartheta^2 \hat{\mathbb{E}}_{y, y' \in \mathcal{X}_{\xi_k}} [|z_k(y) - z_k(y')|].$$
  
23    end  
    // (v) validation set update  
24    Form the trained and identity pairs as in step (v);  
25    Keep the larger-ESS pair as  $(\tilde{\mathcal{Y}}_k, w_k)$  and set  $G_k$  to the corresponding map;  
26    if  $\text{ESS}(w_k) < \tau_v$  then  
      // reject the stage and shrink toward  $t_{k-1}$   
27       $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$ , then update  $U_{t_k} \leftarrow (1 - t_k)U_0 + t_kU$ ;  
28    else  
29       $\mathcal{Y}_k \leftarrow \text{MALA}(\text{resample}(\tilde{\mathcal{Y}}_k, w_k; N), U_{t_k})$ ;  
30    end  
31  until  $\text{ESS}(w_k) \geq \tau_v$ ;  
32 end  
33  $K \leftarrow k$ ;  
34 return  $\{t_k\}_{k=0}^K$  and  $\{G_k\}_{k=1}^K$ ;
```

4.4 Propagation factor analysis

Algorithm 4 returns a validation set of N samples at the endpoint. This subsection bounds the Monte Carlo error of that validation set as an estimate of the target distribution μ and identifies how the bound grows along the stage schedule. We analyze the flow and resample part of the generator, omitting the MALA rejuvenation for simplicity. The stage maps are the diffeomorphisms fixed by an accepted schedule; no property of the training that produced them is used.

Assuming $\mu_k \ll \nu_k$, we take the stage weight in normalized form, $w_k := d\mu_k/d\nu_k$ with $\int w_k d\nu_k = 1$, and assume $w_k > 0$ so that the self-normalization below is always defined. The scheme draws $X_0^1, \dots, X_0^N$ independently from μ_0 and, for $k = 1, \dots, K$, repeats three steps.

- *Propagation.* Apply the inverse stage map to each particle, $Y_k^i = G_k^{-1}(X_{k-1}^i)$.
- *Reweighting.* Evaluate the stage weight at each propagated particle, $W_k^i = w_k(Y_k^i)$.
- *Resampling.* Draw $X_k^{1:N}$ by multinomial resampling from $\sum_{i=1}^N (W_k^i / \sum_{l=1}^N W_k^l) \delta_{Y_k^i}$.

After stage K the scheme returns the empirical measure $\mu_K^N = \frac{1}{N} \sum_{i=1}^N \delta_{X_K^i}$ as the estimate of μ_K . This is a simplified model of inference with a trained generator: every G_k is fixed, the accepted schedule is replayed on a fresh particle set with no optimizer update, and the MALA rejuvenation that Algorithm 4 applies after each resampling is dropped. No Markov kernel therefore acts between stages, so the error bounded here is the propagation error alone. For a distribution μ and a function φ , $\mu(\varphi) := \int_{\mathbb{R}^d} \varphi d\mu$ is the integral, or average, of φ under μ . For a bounded observable φ write $\text{osc}(\varphi) = \text{ess sup } \varphi - \text{ess inf } \varphi$, and for a function h write $\|h\|_\infty$ for its essential supremum on $\mathbb{R}^d$, taken for w_k with respect to ν_k . Since $w_k > 0$ and every G_k is a diffeomorphism, the distributions $\mu_0, \nu_1, \dots, \nu_K$ and $\mu_1, \dots, \mu_K$ are mutually equivalent, so the null sets these quantities ignore are the same for all of them.

Theorem 2 *Suppose $w_k > 0$ and the ν_k -essential supremum $\|w_k\|_\infty$ is finite, for every accepted stage k . Then for every bounded observable φ and every $N \geq 1$,*

$$\mathbb{E} \left[(\mu_K^N(\varphi) - \mu_K(\varphi))^2 \right] \leq \frac{C}{N}, \quad C = \frac{\text{osc}(\varphi)^2}{4} \left(\sum_{k=0}^K \prod_{j=k+1}^K 3\|w_j\|_\infty \right)^2, \quad (23)$$

with the empty product equal to one.

Appendix B gives the proof of Theorem 2. The constant is explicit and independent of N , and depends on the stage maps only through the weight suprema; it is a product over stages, summed over the stage at which the error enters.

Theorem 2 also settles how the scheme behaves in N . The bound gives $\|\mu_K^N(\varphi) - \mu_K(\varphi)\|_2 \leq \sqrt{C/N}$ for every bounded observable, which also bounds the bias and vanishes as N grows: the flow and resample inference is *asymptotically unbiased*, and its error is removed by increasing the sample size alone. Classical Monte Carlo sampling of μ instead requires the simulation time to grow and, for unadjusted integrators, the time step to vanish; a Metropolis-adjusted kernel such as MALA removes the second requirement but not the first. The generator of Algorithm 4 inherits the property, since its rejuvenation step is MALA and leaves each stage target invariant exactly (Del Moral, 2004). That step is also what keeps the particles distinct. Multinomial resampling only copies particles, and the stage maps are bijections, so flow and resample alone can never raise the number of distinct configurations: it can only fall from stage to stage, and the returned set is made increasingly of repeats. The MALA sweep after each resampling diffuses the copies apart under independent noise, so the particles carried into the next stage are scattered configurations rather than duplicated ones, and because the kernel is μ_k -invariant this costs no bias.

The structure of that constant in (23) motivates two propagation factors, which characterize how the L^2 error propagates across the stages. The first is the square root of the constant, up to the observable,

$$F_{\Sigma+} := \sum_{k=0}^K \prod_{j=k+1}^K 3\|w_j\|_\infty, \quad (24)$$

so that $C = \frac{1}{4} \text{osc}(\varphi)^2 F_{\Sigma+}^2$, and (23) reads

$$\|\mu_K^N(\varphi) - \mu_K(\varphi)\|_2 \leq \frac{\text{osc}(\varphi)}{2} \frac{F_{\Sigma+}}{\sqrt{N}}.$$

The factor therefore acts on the error itself, and by Theorem 2 the L^2 error decays at rate $N^{-1/2}$, the same rate as the Monte Carlo error.

The second keeps that form, drops the factors 3, which arise from the recursion in the proof and carry no information about the problem, and replaces each supremum by the reciprocal square root of the stage ESS. For the stage weight, the ESS of (19) reads

$$\text{ESS}(w_k) := \frac{1}{\int_{\mathbb{R}^d} w_k(x)^2 \nu_k(x) dx} \in (0, 1], \quad (25)$$

and the resulting factor is

$$F_{\Sigma} := \sum_{k=0}^K \prod_{j=k+1}^K \frac{1}{\sqrt{\text{ESS}(w_j)}}. \quad (26)$$

The substitution is ordered stagewise. Since $\|w_j\|_{\infty} \geq \int w_j d\nu_j = 1$,

$$\frac{1}{\text{ESS}(w_j)} = \int_{\mathbb{R}^d} w_j^2 d\nu_j \leq \|w_j\|_{\infty} \leq \|w_j\|_{\infty}^2 \quad \text{for every accepted stage } j \implies F_{\Sigma} \leq F_{\Sigma+}.$$

Neither factor can be computed directly: F_{Σ} requires the ν_j -integral $\int w_j^2 d\nu_j$, and $F_{\Sigma+}$ the ν_j -essential supremum of w_j . A Boltzmann generator run already evaluates the importance weights of every stage, $\mathbf{W}_j = (W_j^1, \dots, W_j^N)$ with $W_j^i = w_j(Y_j^i)$, and the sample counterparts of both quantities are read off them at no further cost. We therefore propose the discrete approximations of the factors

$$\hat{F}_{\Sigma+} := \sum_{k=0}^K \prod_{j=k+1}^K \max_{1 \leq i \leq N} \frac{3NW_j^i}{\sum_{l=1}^N W_j^l}, \quad \hat{F}_{\Sigma} := \sum_{k=0}^K \prod_{j=k+1}^K \frac{1}{\sqrt{\text{ESS}(\mathbf{W}_j)}}, \quad (27)$$

both invariant to the scale of the weights and therefore available when the target is known only up to a constant. They are ordered by the same three inequalities, with the largest rescaled weight in place of the supremum, so $\hat{F}_{\Sigma} \leq \hat{F}_{\Sigma+}$. Neither, however, is a bound on its continuous counterpart at finite N .

The experiments report $\hat{F}_{\Sigma}$ as the propagation factor, while the analysis stays with $F_{\Sigma+}$ and F_{Σ} . F_{Σ} is not itself a bound on the error constant; it is a realistic and convenient surrogate for $F_{\Sigma+}$, built directly from the ESS of each accepted stage. Every summand is at least one, so $F_{\Sigma} \geq K + 1$, and the cost of resampling is charged even when every stage proposal equals its target. Appendix B.3 takes up what this substitution preserves and what it does not: why F_{Σ} is not an upper bound on the error constant, how far it can sit from $F_{\Sigma+}$, and why it is nevertheless the quantity the experiments report.

The factor summarizes weight degeneracy along one accepted schedule, and does not measure the difficulty of the target. The sample size required by importance sampling grows exponentially in the KL divergence between target and proposal (Chatterjee and Diaconis, 2018), and with the intrinsic dimension of the problem (Agapiou et al., 2017); neither dependence appears in F_{Σ} .

5 Numerical experiments on model problems

We test forward KL against its X-augmented losses $\text{KL} + \text{X}_{\mu}$, $\text{KL} + \text{X}_{\mu} + \text{X}_{\hat{\mu}}$, and $\text{KL} + \text{X}_{\mu} + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$, the last of which is KLXX. The tests are the 2D benchmark distributions, the high-dimensional product multi-well, the lattice ϕ^4 field theory, and the p -state clock model. They vary in dimension, in the kind of barrier that separates the modes, and in how far those modes lie from the source. Where a fake ESS can be an issue, we report ESS together with the coverage diagnostic.

5.1 2D benchmark distributions

For every target, we initialize the same neural spline flow (NSF) (Durkan et al., 2019) at the identity and compare the four losses above. The flow has 32 spline bins, 6 transforms, and conditioner widths (128, 128), and each training step constructs a target surrogate by one annealing update through the current flow, followed by MALA rejuvenation. For both losses containing the wide-coverage term $X_{\hat{\mu}}$, QT acts on a fixed validation set of 5×10^4 to 8×10^4 points, and the final ESS is computed from flow importance weights on that set. Coverage, with $\kappa = 5$ against an independently generated QT reference set, measures the fraction of reference configurations represented by the pushforward samples through the nearest-neighbor test of Section 3.4. Because coverage tests occupancy of the reference balls, a collapsed flow can have ESS near one but cannot have full coverage. Each loss-target pair is one fixed-seed training realization, so the numerical differences below describe these matched runs rather than variation across independent trainings.

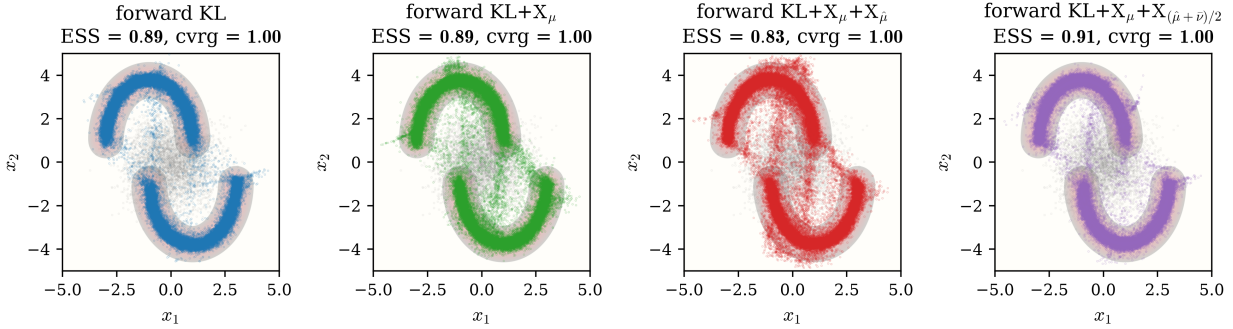


Figure 1: Two-Moon target. Columns, from left to right, are forward KL, KL+ X_{μ} , KL+ X_{μ} + $X_{\hat{\mu}}$, and KL+ X_{μ} + $X_{(\hat{\mu}+\hat{\nu})/2}$. Each panel overlays the pushforward samples on the target energy and reports final ESS and coverage. Gray points show the source.

Two-Moon. The source is $\mathcal{N}(0, I_2)$ and both crescents lie within its reach, so all four losses attain coverage 1. With batch size 500, 200 Adam steps, and learning rate 10^{-3} , forward KL reaches ESS 0.8934. Adding X_{μ} leaves it at 0.8929, while the equal-weight mixture is best at 0.9087 (Figure 1). The loss KL+ X_{μ} + $X_{\hat{\mu}}$ falls to 0.8313, below forward KL, since QT supplies no mode that the annealed batch is missing. Because coverage is already complete, this comparison tests whether pairwise log-ratio control improves density calibration on the reached support.

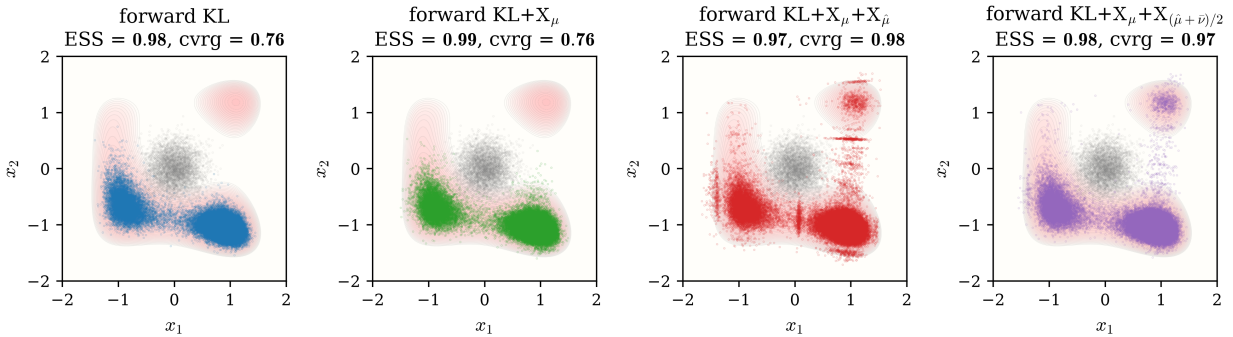


Figure 2: Three-Well target. From left to right, the panels show forward KL, KL+ X_{μ} , KL+ X_{μ} + $X_{\hat{\mu}}$, and KL+ X_{μ} + $X_{(\hat{\mu}+\hat{\nu})/2}$. Each panel overlays the pushforward samples on the target energy and reports final ESS and coverage; gray points show the source.

Three-Well. The source is $\mathcal{N}(0, 0.25^2 I_2)$, narrower than the well separation and initially covering no basin. Training uses batch size 1000, 500 steps, and learning rate 10^{-3} . Forward KL and $\text{KL}+X_\mu$ fill the two favored wells and report ESS 0.9769 and 0.9939, respectively, while their coverage remains 0.7595. The QT set also represents the suppressed third well. Both losses that use it cover all three wells in the sample panels, with scalar coverages 0.9770 and 0.9705 and ESS 0.9680 and 0.9821 (Figure 2).

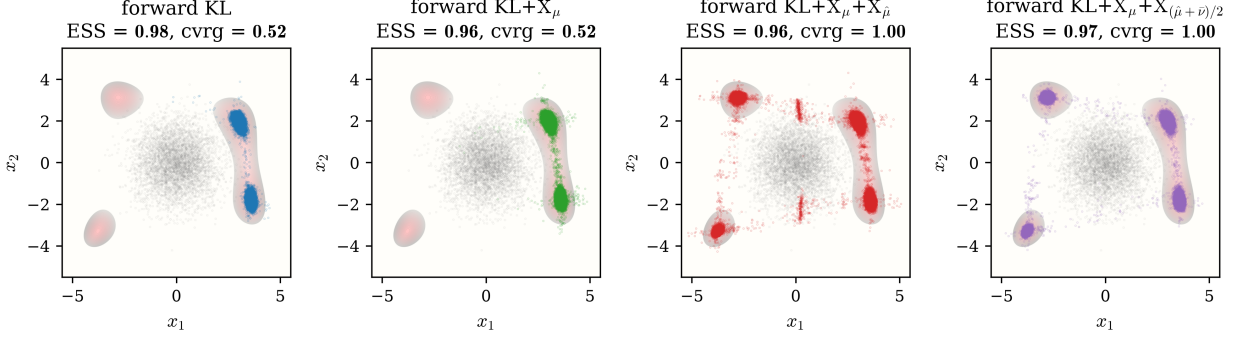


Figure 3: Himmelblau target. Columns show forward KL, $\text{KL}+X_\mu$, $\text{KL}+X_\mu+X_{\hat{\mu}}$, and $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$; each panel shows the pushforward samples over the target energy with final ESS and coverage, and gray points show the source.

Himmelblau. The source is $\mathcal{N}(0, I_2)$, whereas the four wells lie at radius approximately 3.6–5.0. Training uses batch size 1000, 1000 steps, and learning rate 10^{-3} . Forward KL and $\text{KL}+X_\mu$ capture only two wells, giving coverage 0.5160 despite ESS 0.9814 and 0.9596. Both losses using $\hat{\mu}$ attain coverage 1. Relative to the $\text{KL}+X_\mu+X_{\hat{\mu}}$ result, KLXX shows less intermodal mass and raises ESS from 0.9618 to 0.9726 (Figure 3).

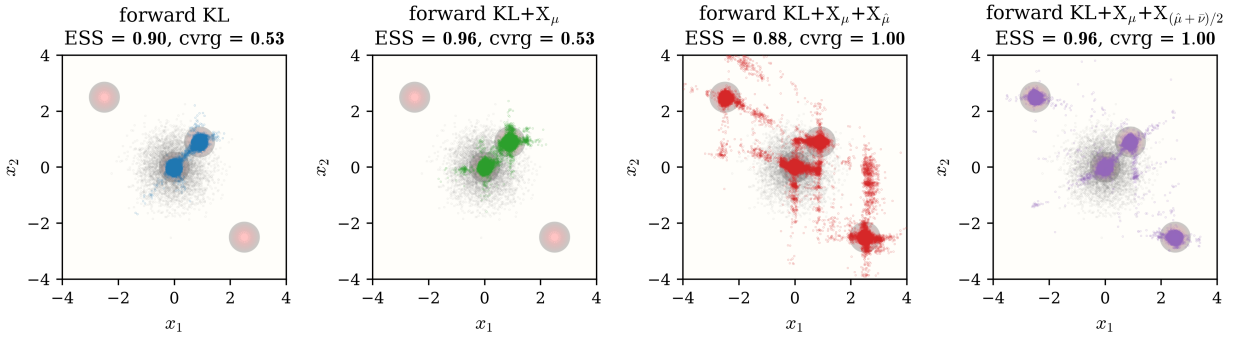


Figure 4: Sparse target. Columns show forward KL, $\text{KL}+X_\mu$, $\text{KL}+X_\mu+X_{\hat{\mu}}$, and $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$, with the pushforward samples overlaid on the target energy; panel titles report final ESS and coverage, and gray points show the source.

Sparse. The target is a four-component Gaussian mixture with component standard deviation 0.08: two modes lie near the source center and two lie in opposite corners, more than five source standard deviations away. With source standard deviation 0.6, batch size 1000, 1000 steps, and learning rate 5×10^{-3} , forward KL and $\text{KL}+X_\mu$ cover only the central pair. Their coverage is 0.5335, while their ESS values are 0.9031 and 0.9567. The loss $\text{KL}+X_\mu+X_{\hat{\mu}}$ restores all four modes but also places visible mass between them, with ESS 0.8783. KLXX attains coverage 1, shows less intermodal mass, and reaches ESS 0.9559.

The four targets support the intended division of labor. Adding X_μ leaves ESS at the forward KL level on Two-Moon, where coverage is already complete. QT samples supply information about modes absent

target	KL	KL+ X_μ	KL+ $X_\mu+X_{\hat{\mu}}$	KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$
Two-Moon	0.893 (1.000)	0.893 (1.000)	0.831 (1.000)	0.909 (1.000)
Three-Well	0.977 (0.760)	0.994 (0.760)	0.968 (0.977)	0.982 (0.971)
Himmelblau	0.981 (0.516)	0.960 (0.516)	0.962 (1.000)	0.973 (1.000)
Sparse	0.903 (0.534)	0.957 (0.534)	0.878 (1.000)	0.956 (1.000)

Table 2: Final ESS, with $\kappa = 5$ coverage in parentheses, on the four 2D targets. Bold marks the highest ESS among methods whose sample panel covers every target mode. The scalar coverage diagnostic slightly undercounts the visibly occupied third mode on Three-Well.

from the AIS samples. Including pushforward sample associated with cleaner intermodal regions than the loss KL+ $X_\mu+X_{\hat{\mu}}$. KLXX has the highest mode-complete ESS on every target in Table 2.

5.2 High-dimensional product multi-well

We next isolate accuracy scaling over the dimensions $d = 16, 32, 48, \dots, 256$. Letting $k = \lfloor \log_2 d \rfloor$, we use

$$U(\mathbf{x}) = \frac{1}{2} \|\mathbf{x}\|^2 + 12 \sum_{i=1}^k \exp(-x_i^2). \quad (28)$$

The first k coordinates are symmetric double wells with minima at $\pm\sqrt{\ln 24}$ and barrier approximately 9.9, while the remaining $d - k$ coordinates are standard Gaussian. The target therefore has exactly 2^k equal-weight sign-pattern modes, where $2^k \leq d < 2^{k+1}$. Multimodality occupies only a k -dimensional subspace, but the log-weight error accumulates across all d coordinates.

All four losses use the same identity-initialized NSF on $[-4, 4]^d$, with 16 bins, 6 transforms, and conditioner widths (256, 256). At every dimension, training uses batch size 250, 2000 Adam steps at learning rate 10^{-3} , and a fixed validation set of size $5000d$. The live target surrogate uses one annealing update followed by 50 MALA steps of size 2×10^{-3} . QT uses $1000d$ samples, melt scale 2, and an Armijo L-BFGS quench with initial step 0.5 and 200 iterations. Final validation ESS is evaluated on 80000 fresh source samples. Each loss-dimension pair is one fixed-seed training realization.

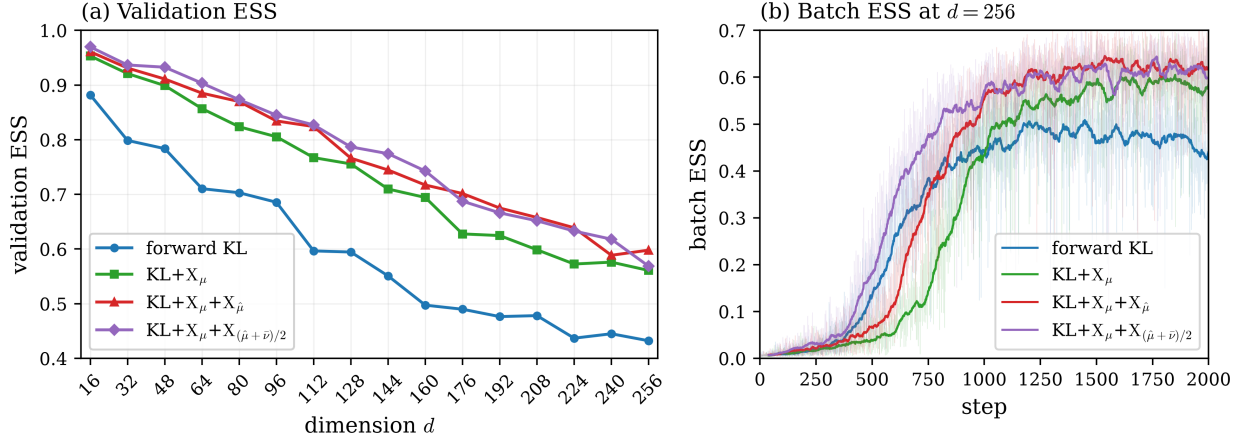


Figure 5: ESS on the product multi-well target (28). (a) Final validation ESS over all sixteen tested dimensions. (b) Batch ESS during the $d = 256$ training run; faint curves are raw values and bold curves are 40-step moving averages.

In each of the 64 evaluated pairs of loss and dimension, all 2^k sign patterns of the double-well coordinates occur among the fresh evaluation samples, including all 256 patterns for each of the four losses at $d = 256$.

Figure 5 shows that each of the three losses with log-ratio variation has higher final validation ESS than forward KL at every one of the sixteen dimensions, so forward KL is the lowest of the four at each dimension. At $d = 256$, the final values are 0.4322 for forward KL, 0.5608 for $\text{KL} + \text{X}_\mu$, 0.5979 for $\text{KL} + \text{X}_\mu + \text{X}_{\hat{\mu}}$, and 0.5687 for $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$. The two losses containing $\hat{\mu}$ behave similarly across dimensions. During the $d = 256$ training realization, the batch ESS of $\text{KL} + \text{X}_\mu$ rises more slowly than that of forward KL over the first several hundred steps and overtakes it only near step 800. Adding $\text{X}_{\hat{\mu}}$ removes that delay: both losses containing $\hat{\mu}$ cross forward KL earlier, the mixture term first, so the wide-coverage term recovers the initial convergence speed and improves on it. All three losses with log-ratio variation then continue to improve after forward KL begins to plateau.

5.3 Lattice ϕ^4 field theory: fake ESS at a collective barrier

On a periodic $L \times L$ lattice with $d = L^2$, the tilted scalar ϕ^4 action is

$$U(\phi) = -0.8 \sum_{\langle j,l \rangle} \phi_j \phi_l + \sum_{j=1}^d \left[\phi_j^2 + \frac{1}{2}(\phi_j^2 - 1)^2 \right] + h \sum_{j=1}^d \phi_j.$$

The on-site term alone has a single minimum at 0; the nearest-neighbor coupling is what makes a uniform configuration bistable, with effective per-site potential $-0.6\phi^2 + \frac{1}{2}(\phi^2 - 1)^2$ and wells near ± 1.26 . The magnetization of a configuration is

$$m(\phi) = \frac{1}{d} \sum_{j=1}^d \phi_j.$$

We take $h = 0.0257$ at $L = 6$ and $h = 0.0144$ at $L = 8$, matching the scaling $hL^2 \approx 0.92$ to produce comparable well imbalance in the targets of the two settings. A collective domain-wall barrier separates the two wells, and the minority weight $p_+ = \mathbb{P}(m > 0)$ must be learned rather than fixed by symmetry.

PT and mirror MALA provide the references $p_+ = 0.1251 \pm 0.0009$ for $L = 6$ and 0.1284 ± 0.0014 for $L = 8$. Each loss is run at seeds 0, 1, 2 using an identity-initialized NSF on $[-3, 3]^d$, with 16 bins, 6 transforms, and conditioner widths (256, 256). The source is $\mathcal{N}(0, 0.5^2 I_d)$, centered on the barrier. Training uses a fixed validation set of 10^5 source samples, batch size 500, and 2000 Adam steps at learning rate 10^{-3} . The target surrogate uses one annealing update followed by 50 MALA steps of size 2×10^{-3} . For both losses containing $\text{X}_{\hat{\mu}}$, quench and temper acts on the validation set, with melt scale 2 and an Armijo L-BFGS quench using initial step 0.1 and 200 iterations.

	$L = 6$		$L = 8$	
	ESS	p_+	ESS	p_+
forward KL	0.794/0.840/0.826	1/1/0	0.622/0.686/0.684	1/0/1
$\text{KL} + \text{X}_\mu$	0.935/0.933/0.945	1/1/0	0.804/0.824/0.821	1/0/0
$\text{KL} + \text{X}_\mu + \text{X}_{\hat{\mu}}$	0.875/0.873/0.871	0.127/0.128/0.127	0.636/0.645/0.654	0.125/0.124/0.124
$\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$	0.896/0.877/0.894	0.127/0.128/0.127	0.690/0.676/0.681	0.131/0.130/0.129
PT reference	—	0.1251 ± 0.0009	—	0.1284 ± 0.0014

Table 3: Final ESS and reweighted minority-well weight p_+ for three seeds on the two tilted ϕ^4 targets, $(L, h) = (6, 0.0257)$ and $(8, 0.0144)$. Integer values 0 and 1 indicate collapse onto one well; bold marks the higher ESS among the losses that recover both wells.

Forward KL and $\text{KL} + \text{X}_\mu$ collapse at every size and seed. Their annealed batches start from pushforward samples and do not cross the collective barrier within the run, so the second well remains absent and reweighting gives $p_+ = 0$ or 1. Nevertheless, $\text{KL} + \text{X}_\mu$ has the highest ESS at both sizes, and it is a fake ESS: it is earned on the single well the flow reaches, and the missing one does not enter it. KLXX has both a trustworthy ESS and samples that populate both wells.

The two KL losses using the QT samples $\hat{\mu}$ populate both wells at every seed. Their recovered p_+ lies between 0.1238 and 0.1312, within 0.005 of the PT estimate and the resampled mirror-MALA reference at

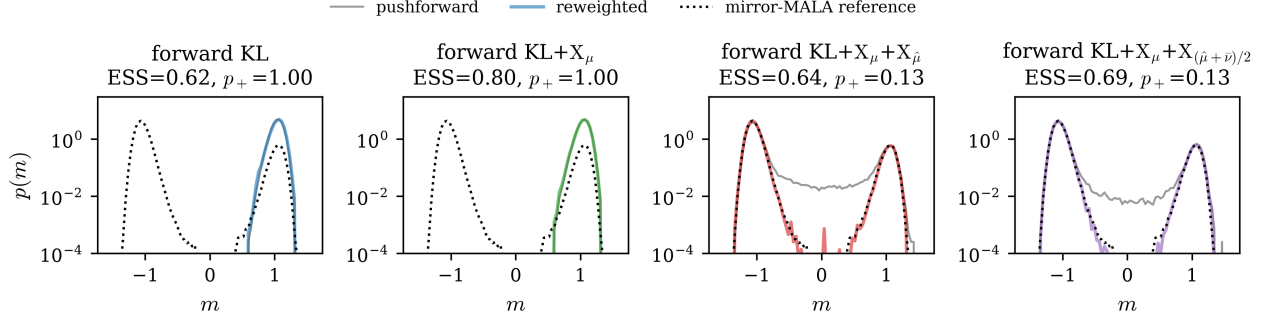


Figure 6: Magnetization density on the $L = 8$ lattice at seed 0. Each panel shows the pushforward density (gray), the reweighted density (color), and the resampled mirror-MALA reference density (dotted); the title reports ESS and the reweighted p_+ .

the corresponding lattice size. In these finite-capacity fits, recovering both wells is accompanied by some mass in the suppressed region between them and lower ESS. At $L = 8$, KLXX has ESS 0.6756–0.6902, compared with 0.6355–0.6541 for the loss $\text{KL} + X_\mu + X_{\hat{\mu}}$. Its samples retain both wells with less visible mass in the suppressed region between them, consistent with the intended role of the pushforward samples.

5.4 Boltzmann generator on p -state clock model

The preceding benchmarks use a single flow. We now test the full adaptive-staging construction on the $p = 6$ clock model on a periodic 8×8 lattice,

$$U(\boldsymbol{\theta}) = - \sum_{\langle jl \rangle} \cos(\theta_j - \theta_l) - \frac{1}{2} \sum_{j=1}^d \cos(6\theta_j), \quad \boldsymbol{\theta} \in [-\pi, \pi)^d, \quad d = 64, \quad (29)$$

where $\langle jl \rangle$ runs over nearest-neighbor pairs. The source is uniform on the torus and the interpolation is $U_t = tU$. The target distribution $\mu \propto \exp(-U)$ separates into six symmetry-related clock sectors.

Each stage uses an identity-initialized neural circular spline flow with 16 bins, 6 autoregressive transforms, and conditioner widths (256, 256). The validation set contains 4×10^5 samples, and both SMC and QT act directly on this set. Every stage uses 3000 Adam steps at learning rate 10^{-3} and a 4-level annealed surrogate with 100 MALA steps of size 10^{-3} . For KLXX, QT uses melt scale 2π and an Armijo L-BFGS quench with initial step 10^{-2} and 100 iterations. The clock stage schedule uses the nondefault settings $t_{\text{safe}} = 0.25$, enlarge factor $\Gamma = 2$, and gates $\tau_{\text{SMC}} = 0.7$ and $\tau_v = 0.4$. We report batch sizes $B = 2000, 1000, 500, 250$.

KLXX uses the same staging schedule with forward KL for fair comparison. That schedule is constructed with forward KL under SMC preselection and an ESS acceptance gate, and KLXX is then trained on the identical accepted values of t without a second gate. Every configuration is a single run, so individual stage ESS values carry single-run fluctuation.

The principal diagnostic is the validation ESS of each accepted stage. The propagation factor $\hat{F}_\Sigma$ of (27) summarizes stagewise weight degeneracy across the stage sequence; smaller is better. It is not a full-chain ESS or an endpoint error estimate. Table 4 reports every accepted stage and the resulting factor.

KLXX has higher validation ESS on all 28 shared stages, raising the geometric mean stage ESS from 0.556 to 0.665 at $B = 2000$, from 0.553 to 0.643 at $B = 1000$, from 0.487 to 0.571 at $B = 500$, and from 0.470 to 0.553 at $B = 250$. It reduces the propagation factor $\hat{F}_\Sigma$ at every reported batch size, by ratios between 1.33 and 1.50: the factor falls from 21.4 to 15.6 at the largest batch and from 62.7 to 41.7 at the smallest. As the batch shrinks, the schedule lengthens and both propagation factors grow.

The single-site marginal exhibits all six clock wells with nearly equal peak heights. Pair differences are most concentrated at zero for nearest neighbors; as the lattice separation increases, the alignment peak falls and the shoulders near $\pm\pi/3$ gain mass. This is the expected decay of local angular order with distance.

To quantify whether staged resampling preserves the sixfold symmetry, we perform two fresh-rebuild tests through the trained stage sequences. The batch size test fixes $N = 6.4 \times 10^5$, uses $B \in \{2000, 1000, 500, 250\}$,

stage k	1	2	3	4	5	6	7	8	$\hat{F}_\Sigma$
t_k ($B = 2000$)	0.250	0.495	0.663	0.779	0.887	1.000	—	—	
forward KL	0.751	0.570	0.471	0.474	0.511	0.602	—	—	21.4
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.921	0.759	0.608	0.562	0.558	0.649	—	—	15.6
t_k ($B = 1000$)	0.250	0.495	0.613	0.728	0.841	0.952	1.000	—	
forward KL	0.672	0.479	0.580	0.463	0.427	0.520	0.830	—	28.3
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.866	0.642	0.700	0.545	0.462	0.541	0.859	—	21.2
t_k ($B = 500$)	0.250	0.421	0.590	0.705	0.784	0.895	1.000	—	
forward KL	0.580	0.549	0.408	0.432	0.525	0.417	0.530	—	40.7
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.778	0.695	0.506	0.500	0.595	0.429	0.566	—	29.3
t_k ($B = 250$)	0.250	0.421	0.539	0.654	0.734	0.811	0.887	1.000	
forward KL	0.468	0.448	0.495	0.428	0.506	0.484	0.514	0.424	62.7
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.659	0.586	0.601	0.480	0.563	0.536	0.564	0.464	41.7

Table 4: Selected-proposal validation ESS on every accepted stage of the schedule-matched clock-model comparison between forward KL and KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$. Within each batch size block, both losses traverse the listed stage schedule. The final column is the propagation factor $\hat{F}_\Sigma$ in (27); bold marks the better loss at each stage and the smaller propagation factor.

and averages over four matched sampling seeds for each method and batch size. The particle-count test fixes $B = 2000$ and uses $N = 10^4 2^k$, $k = 0, \dots, 6$, with 2^{8-k} independent rebuilds at each size so every particle-count value has equal total work of 2.56×10^6 particles. For each sample, let s be the clock sector nearest to the argument of the complex magnetization $d^{-1} \sum_{j=1}^d e^{i\theta_j}$, and let p_s be the empirical fraction assigned to sector s . We define

$$\text{occupancy bias} := \frac{1}{6} \sum_{s=0}^5 \left| p_s - \frac{1}{6} \right|.$$

At fixed N , panel (a) shows that the mean occupancy bias increases as the training batch shrinks for both methods. KLXX has the lower mean at $B = 2000$, 1000, and 250. At those batch sizes, the forward KL means are 0.00573, 0.00888, and 0.02760, while the KLXX means are 0.00235, 0.00517, and 0.01506. At $B = 500$ the ordering reverses slightly, from 0.00963 for forward KL to 0.01082 for KLXX, and the intervals defined by ± 2 standard errors overlap.

At fixed $B = 2000$, panel (b) follows the Monte Carlo rate: the log-log slopes of occupancy bias against N are -0.459 for forward KL and -0.479 for KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$. The absence of a visible bias floor is consistent with sampling fluctuation rather than a particle-count-independent sector imbalance over the tested range. The KLXX curve is lower at all seven reported sizes, reducing the mean bias from 0.03735 to 0.02397 at $N = 10^4$ and from 0.00552 to 0.00358 at $N = 6.4 \times 10^5$. Together with the six peaks in Figure 7, these tests show that the staged samples are not confined to one global sector. The rebuild seeds measure sampling variability; each stage schedule for a given method and batch size is one training run, so the between-method differences do not estimate variation over independently trained models.

6 Boltzmann generators on molecular potential energies

The model problems above separate the target and QT terms on explicit targets and probe the term built from pushforward samples. Molecular systems add a singular nonbonded core and more conformational degrees of freedom as molecular size grows, and their whitened coordinates (Noé et al., 2019) bring a mixed Euclidean-periodic domain. We first define the molecular measure and its regularized potential, then add sharpening to the staged generator and report the completed alkane family, three additional achiral molecules, and three chiral KLXX runs with fixed stereochemical configurations. The reported molecular wall times were recorded on a single NVIDIA GeForce RTX 5090 GPU.

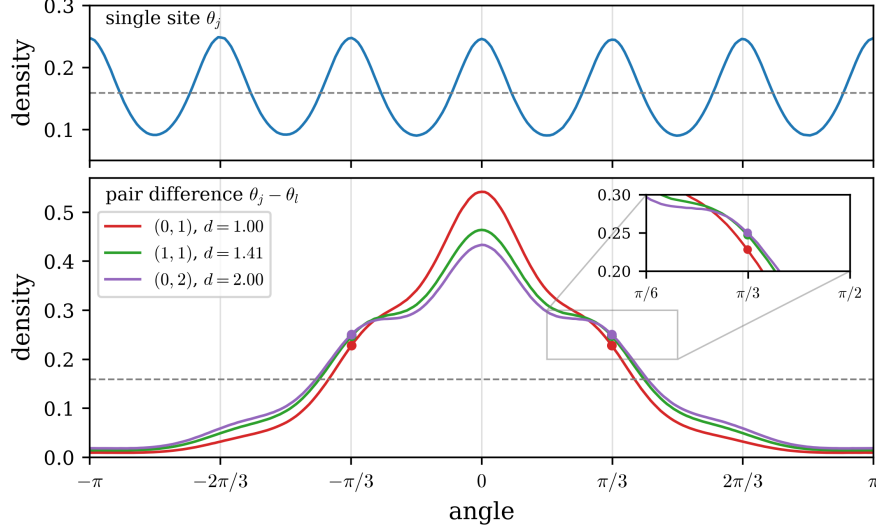


Figure 7: Angle densities from 2×10^6 fresh samples advanced through the trained $B = 2000$ $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$ stage sequence by map, reweighting, resampling, and MALA. The upper panel is the single-site marginal. The lower panel gives pair differences at offsets $(0, 1)$, $(1, 1)$, and $(0, 2)$; dots and the inset resolve the one-clock-step angle $\pi/3$. Dashed lines show the uniform density.

6.1 Molecular potential models

For Cartesian coordinates x , let $E(x)$ denote the complete molecular-mechanics energy

$$E(x) = E_{\text{bond}} + E_{\text{angle}} + E_{\text{torsion}} + E_{\text{LJ}} + E_{\text{Coul}} + E_{\text{GB}}^{\text{OBC1}} + E_{\text{ACE}}. \quad (30)$$

The bonded and vacuum nonbonded part has the Amber form (Cornell et al., 1995)

$$\begin{aligned} E_{\text{vac}}(x) = & \sum_b k_b (r_b - r_b^0)^2 + \sum_a k_a (\theta_a - \theta_a^0)^2 + \sum_\tau \frac{V_\tau}{2} [1 + \cos(n_\tau \varphi_\tau - \gamma_\tau)] \\ & + \sum_{\substack{i < j \\ \text{nonbonded}}} \left\{ 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + k_e \frac{q_i q_j}{r_{ij}} \right\}, \end{aligned} \quad (31)$$

Here b , a , and τ index bonds, angles, and proper or improper torsions, while i, j index nonbonded atom pairs after topology-defined exclusions and scaling. The symbols r_b^0 , θ_a^0 , k_b , k_a , V_τ , n_τ , and γ_τ are force-field equilibrium values and coefficients; σ_{ij} , ϵ_{ij} , and q_i are the nonbonded parameters. OBC1 supplies the generalized-Born electrostatic solvation term (Onufriev et al., 2004), and ACE supplies the nonpolar term (Schaefer and Karplus, 1996). All terms in (30) belong to the sampled potential.

A nonlinear molecule with A atoms has $d = 3A - 6$ internal degrees of freedom after rigid translation and rotation are removed. A fixed internal-coordinate tree reconstructs $x = x(q)$ from whitened bond and angle coordinates and periodic torsions. Write d_E and d_T for the numbers of nonperiodic Euclidean coordinates and periodic torsions, respectively, and let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. Consequently,

$$\mathcal{Q} = \mathbb{R}^{d_E} \times \mathbb{T}^{d_T}, \quad d_E + d_T = 3A - 6, \quad (32)$$

and the physical reduced potential at inverse temperature $\beta_T = (k_B T)^{-1}$ is

$$U(q) = \beta_T E(x(q)) - \log J(q), \quad (33)$$

where $J(q)$ is the Cartesian volume Jacobian of the reconstruction, including the whitening maps. The source is Gaussian on the Euclidean block and uniform on the torus. The mixed spline flow uses ordinary rational-quadratic splines on the Euclidean coordinates and circular splines on the torsions.

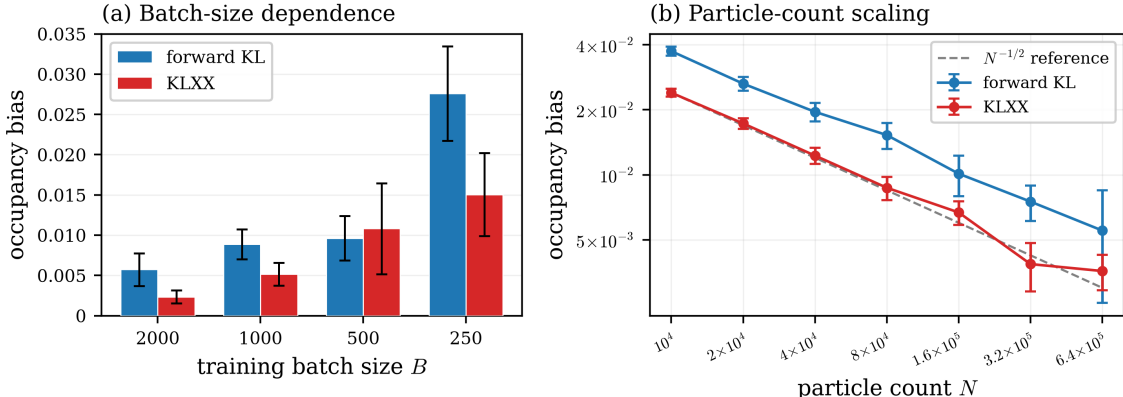


Figure 8: Clock-sector occupancy bias under two fresh-rebuild tests. (a) Mean bias at fixed $N = 6.4 \times 10^5$ across training batch sizes; each bar averages four matched sampling seeds. (b) Mean bias at fixed $B = 2000$ versus particle count; at $N = 2^k \times 10^4$, each mean uses 2^{8-k} independent rebuilds, keeping the total work per point at 2.56×10^6 particles. Error bars in both panels are ± 2 standard errors. The dashed line in (b) is an $N^{-1/2}$ reference.

target	atoms / d	model
<i>Achiral targets</i>		
alkanes, C_1 – C_6	5–20/9–54	GAFF2/AM1-BCC; OBC1/ACE
<i>N</i> -methylacetamide (NMA)	12/30	ff96; OBC1/ACE
glycerol	14/36	GAFF2/AM1-BCC; OBC1/ACE
neutral diethanolamine	18/48	GAFF2/AM1-BCC; OBC1/ACE
<i>Chiral targets: one fixed stereochemical configuration</i>		
alanine dipeptide	22/60	ff96; OBC1/ACE
(2 <i>R</i> , 3 <i>R</i>)-2,3-butanediol	16/42	GAFF2/AM1-BCC; OBC1/ACE
Ac-Pro-NHMe	26/72	ff96; OBC1/ACE

Table 5: Molecular targets in mixed internal coordinates, grouped by configurational stereochemistry. All systems are at 300 K, use no nonbonded cutoff or bond constraints, and include the coordinate Jacobian in (33). For each chiral target, the generator and reference are restricted to one specified stereochemical configuration, while all conformations within that configuration remain part of the target.

Table 5 gives the force field and implicit-solvent model of each target. GAFF and AM1-BCC are described in Wang et al. (2004) and Jakalian et al. (2002), the bundles are prepared with AmberTools (Case et al., 2023), and the independent references are propagated with OpenMM (Eastman et al., 2017). We abbreviate *N*-acetyl-L-proline *N*-methylamide as Ac-Pro-NHMe. The chiral targets retain only the named configurations: L-alanine for alanine dipeptide, (2*R*, 3*R*) for 2,3-butanediol, and L-proline for Ac-Pro-NHMe. The achiral targets fix no stereocenter, and every conformational basin within a selected configuration belongs to the target, so the restriction defines the target rather than filtering its samples.

The Lennard–Jones core in (31) grows as r^{-12} , so early flow proposals can place a few samples at energies and gradients far beyond the scale of the thermally occupied region. We therefore use a two-parameter regularized potential. Let x_* be the bundle reference geometry. After flooring only the regular and exception Coulomb/Lennard–Jones pair distances at r , denote the complete force-field energy by $E_r(x)$, its reference value by $E_r^* := E_r(x_*)$, and its reference-relative value by $\Delta E_r(x) := E_r(x) - E_r^*$. For an energy threshold e , define the soft cap

$$C_e(\Delta E) = \begin{cases} \Delta E, & \Delta E \leq e, \\ e[1 + \log(\Delta E/e)], & \Delta E > e, \end{cases} \quad (34)$$

and collect the two regularization parameters in $\rho = (e, r)$. The corresponding reduced potential is

$$U^{\{\rho\}}(q) = \beta_T [E_r^* + \mathcal{C}_e(\Delta E_r(x(q)))] - \log J(q). \quad (35)$$

The threshold e is measured in kJ mol^{-1} , and the pair floor r is measured in nm. The component formulas of the bonded and OBC1/ACE terms are unchanged before the scalar cap is applied to the complete energy. The coordinate Jacobian and domain are not capped. The physical potential U in (33) is recovered in the formal limit $e \rightarrow \infty$, $r \rightarrow 0$.

A run ends at a finite regularization setting ρ rather than at that formal limit. For endpoint samples $q_1, \dots, q_N$, define the endpoint-to-physical weights and the corresponding regularization ESS (RESS) by

$$w_i^{\text{phys}} := \exp[U^{\{\rho\}}(q_i) - U(q_i)], \quad \text{RESS} := \text{ESS}(w^{\text{phys}}). \quad (36)$$

6.2 Boltzmann generators with sharpening

A softer molecular potential can make early stages easier in a given adaptive-staging run. After each flow stage, *sharpening* reweights the validation set toward a stricter regularization setting, resamples it, and applies MALA under the stricter potential. To extend the source-to-target interpolation of Section 4, let $\rho_0 = (e_0, r_0)$ and $\rho_1 = (e_1, r_1)$ be the soft and final settings and interpolate them by

$$\rho_s = (1 - s)\rho_0 + s\rho_1, \quad s \in [0, 1]. \quad (37)$$

A fixed-regularization run has $\rho_0 = \rho_1$. Extending the interpolation of Section 4 by the single index s , set

$$U_{t,s}(q) = (1 - t)U_0(q) + tU^{\{\rho_s\}}(q), \quad \mu_{t,s} \propto \exp(-U_{t,s}). \quad (38)$$

Thus t remains the source-to-target interpolation parameter defined in Section 4, while s changes only the regularization setting. If s is fixed, suppressing the second index recovers the notation U_t, μ_t of Section 4.

Accepted sample sets lie on the diagonal $s = t$, but one stage advances the two coordinates in sequence:

$$\mathcal{Y}_{k-1} \approx \mu_{t_{k-1}, t_{k-1}} \xrightarrow[\text{selected } (\tilde{\mathcal{Y}}_k, w_k)]{\text{flow or identity}} \mathcal{Y}_k^- \approx \mu_{t_k, t_{k-1}} \xrightarrow[w_k^{\text{sharp}}]{\text{sharpen}} \mathcal{Y}_k \approx \mu_{t_k, t_k}. \quad (39)$$

The first arrow of (39) holds $s = t_{k-1}$ fixed and advances t from t_{k-1} to t_k . Potential-space SMC and mixed-domain MALA carry a copy of all N particles along this bridge, and in KLXX the additional QT samples are built under $U_{t_k, t_{k-1}}$. After training, the candidate flow and identity proposals are evaluated on the carried set $\mathcal{Y}_{k-1}$; the pair with larger validation ESS is denoted by $(\tilde{\mathcal{Y}}_k, w_k)$, ties retaining the identity, and reweighting, resampling, and MALA under $U_{t_k, t_{k-1}}$ produce $\mathcal{Y}_k^-$. Both ends of this arrow carry the same regularization parameters, so the map is left only the interpolation coordinate to absorb.

The second arrow of (39) holds $t = t_k$ fixed and advances the regularization coordinate from t_{k-1} to t_k . Its exact log weight is

$$\log w_k^{\text{sharp}}(q) = U_{t_k, t_{k-1}}(q) - U_{t_k, t_k}(q) = t_k [U^{\{\rho_{t_{k-1}}\}}(q) - U^{\{\rho_{t_k}\}}(q)]. \quad (40)$$

Sharpening is accepted only when its ESS over the validation set clears the stage threshold, after which the samples are resampled and rejuvenated by MALA under U_{t_k, t_k} ; a failed proposal or sharpening gate shrinks $t_k - t_{k-1}$ and retries. This arrow therefore carries the change of regularization by exact reweighting rather than by the map, and it raises barriers the map is not asked to raise: a softer core flattens the short-range repulsion and the high-energy tail, which can leave passages between modes that the endpoint potential does not have, and advancing s toward t_k removes those spurious connections.

The propagation factor of (27) is evaluated on both arrows of (39). An accepted stage on a sharpening path reweights and resamples twice, so it supplies two of the weight vectors $\mathbf{W}_j$: the selected proposal weight w_k of the first arrow and the sharpening weight w_k^{sharp} of (40), each evaluated on the N validation samples. A run of K accepted stages therefore supplies $2K$ weight vectors, K from the proposals and K from sharpening, and the sum and products of $\hat{F}_\Sigma$ run over them in the order the generator performs them. On a fixed regularization path the sharpening weight is identically one, and the step still enters the sequence

because the generator resamples there. As before, $\hat{F}_\Sigma$ summarizes stagewise weight degeneracy; it is not a fresh source-to-endpoint ESS or an endpoint error estimator.

Temperature-annealed Boltzmann generators (TA-BG) (Schopmans and Friederich, 2025) address the same barrier problem through a different continuation coordinate: TA-BG lowers the temperature toward the physical target, whereas sharpening holds the temperature fixed and anneals the regularization ρ_s . Sharpening is the stronger intervention at the singular core, since changing temperature rescales the physical energy but preserves its divergent short-range form. It is also the less interpretable one: before the endpoint the sharpening states are algorithmic ensembles rather than physical molecular systems, whereas the TA-BG intermediates remain canonical distributions of the same potential at explicitly defined elevated temperatures. We have no matched numerical comparison with TA-BG and leave it to future work.

The generators below differ only in how each stage proposal is obtained, and every one of them retains the same SMC selection, reweighting, resampling, and MALA steps.

6.3 Alkane family results

We test methane (CH_4) through n -hexane (C_6H_{14} , straight chain). The regularization paths are listed in Table 6: fixed-regularization runs start and remain at the common endpoint, while sharpening runs begin at a softer setting and advance linearly to it. The three methods use method-specific adaptive stage schedules, so stage counts and recorded wall times are descriptive rather than fixed-compute comparisons.

All molecular flows use 6 mixed spline transforms, 32 bins, conditioner widths (256, 256), spline bound 8, and minimum slope 10^{-3} , and are initialized at the identity. The validation and batch sizes (N, B) are (100000, 5000) for methane, (120000, 6000) for ethane, (160000, 8000) for propane, and (200000, 10000) for butane through hexane. Each accepted flow stage uses 500 Adam steps for methane through propane, 400 for butane, and 300 for pentane and hexane, with learning rate 10^{-3} and 25 warmup steps. Potential-space SMC uses 8 annealing levels; mixed-domain MALA uses 100 steps of size 10^{-3} . QT uses melt scale 1, 200 L-BFGS steps with initial step 10^{-2} , and the complete molecular QT set. The safe start is 0.25 for methane and ethane and 0.1 for the larger molecules. The validation gate is 0.4 throughout; the SMC gate is 0.3 for methane and ethane, 0.5 for propane, and 0.8 for butane through hexane.

molecule (d)	start ρ_0	ID			KL			KLXX		
		$\hat{F}_\Sigma$	stages	time	$\hat{F}_\Sigma$	stages	time	$\hat{F}_\Sigma$	stages	time
methane (9)	(100, 0.15)	26.13	8	1.4	7.464	3	1.2	7.033	3	2.4
ethane (18)	(100, 0.15)	50.02	9	3.5	10.36	3	1.8	7.305	3	4.1
propane (27)	(100, 0.15)	93.24	11	7.7	92.97	11	13.4	20.01	7	23.0
	(50, 0.15)	65.01	10	6.9	64.99	10	12.1	19.41	6	18.3
butane (36)	(100, 0.15)	138.6	12	13.3	138.6	12	20.3	37.49	10	49.0
	(50, 0.20)	86.75	10	10.8	86.68	10	14.9	36.19	10	46.1
pentane (45)	(100, 0.15)	217.0	13	19.2	208.0	13	24.6	65.43	12	60.4
	(50, 0.20)	117.0	10	13.9	117.6	10	17.2	54.89	9	38.5
hexane (54)	(100, 0.15)	334.8	13	23.3	336.5	13	29.0	190.3	12	69.9
	(50, 0.25)	171.3	10	17.4	170.1	10	22.7	114.5	10	60.1

Table 6: Alkane Boltzmann generator training results, including propagation factor $\hat{F}_\Sigma$, the number of accepted stages, and total wall time in minutes. ID applies the identity map at every stage, whereas KL and KLXX may select a trained flow when it improves validation ESS. Each row reports one seed and one regularization path; bold marks the lowest $\hat{F}_\Sigma$ among the three methods for that path. Every path ends at (100, 0.15), and rows that start at (100, 0.15) use no sharpening.

Within the single-seed runs, KLXX has the smallest factor in all ten settings. Forward KL selects trained maps through methane and part of ethane. From propane onward, every accepted KL stage selects the identity proposal, and its factor is close to the ID result. KLXX selects the trained map at every accepted

stage through pentane. At hexane, it selects trained maps in 6 of 12 fixed-path stages and 8 of 10 sharpening stages. The difference between the fixed and sharpening paths increases with molecule size in these runs. The KLXX fixed-path factor divided by the sharpening-path factor is 1.03, 1.04, 1.19, and 1.66 for propane through hexane. The butane, pentane, and hexane KLXX sharpening runs take 3.10, 2.24, and 2.65 times the recorded KL wall time. The KLXX sharpening paths have the lowest factor in the larger-molecule comparisons, with additional QT, reweighting, and flow-training cost.

The endpoint-to-physical weights of (36) are nearly uniform on the final KLXX sample sets: $1 - \text{RESS} = 0$ numerically from methane through butane, 9.5×10^{-6} for pentane, and 2.3×10^{-5} for hexane. These values measure overlap on the sampled sets; they do not prove global equality between $U^{\{\rho_1\}}$ and U .

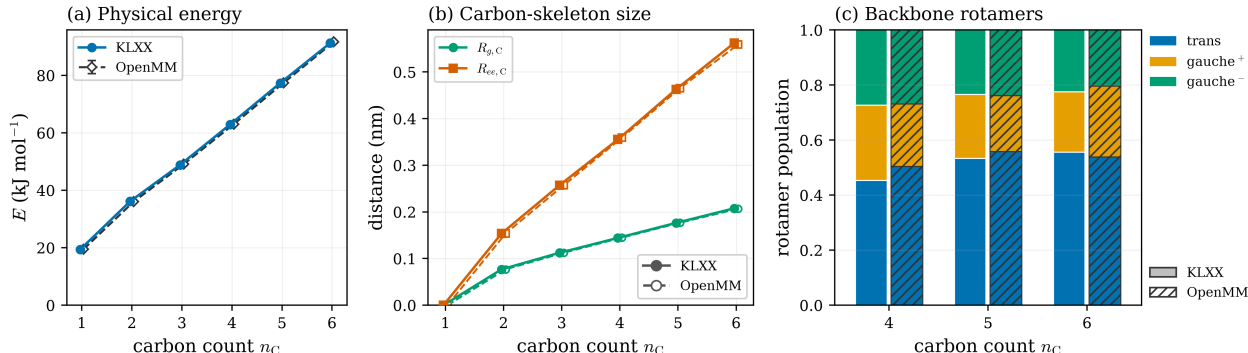


Figure 9: Macroscopic observables from the smallest-factor KLXX sample set for each alkane and an independent physical-potential reference. Fixed-path KLXX is used for methane through propane and sharpening KLXX for butane through hexane. (a) Physical energy evaluated under the exact, unregularized potential. (b) Carbon-skeleton radius of gyration $R_{g,C}$ and terminal-carbon distance $R_{ee,C}$. (c) Pooled backbone trans and gauche fractions. Filled solid curves and unhatched bars denote KLXX; open dashed curves and hatched bars denote the reference.

The KLXX curves are unweighted summaries of the finite-endpoint sample sets; the physical energy E is evaluated only as an observable, without endpoint-to-physical resampling. The physical-potential reference uses two seeds and a 0.25 fs timestep. Methane through propane use 300 K Langevin trajectories; butane through hexane use 300–800 K PT to mix the backbone rotamers. Across the six molecules, the maximum relative difference between KLXX and reference physical-energy means is 0.644%. The largest relative difference among the nonzero carbon radius-of-gyration and terminal-distance means is 1.058%, and the largest absolute rotamer-fraction difference is 0.051. These results show agreement for the selected energy, size, and rotamer summaries across the C_1 – C_6 series.

All thirty ID, KL, and KLXX run manifests used in Table 6 are complete, but each configuration has one training seed. The reference frames are correlated, and the two-seed ranges are descriptive rather than framewise confidence intervals. The table therefore compares the recorded runs; it does not estimate run-to-run variation or distance between the complete equilibrium measures.

6.4 Additional achiral molecule results

We next test NMA, glycerol, and neutral diethanolamine. Each method uses one seed, a validation set of size $N = 200000$, and batch size $B = 10000$. The flow, SMC, MALA, and QT settings match the larger-alkane configuration above. NMA and glycerol use 400 Adam steps per accepted flow stage, while neutral diethanolamine uses 300. The three sharpening paths are listed in Table 7. The two losses use method-specific adaptive stage schedules, so the stage counts and recorded wall times are descriptive rather than fixed-compute comparisons.

Comparing $KL+X_\mu$ with KLXX, the latter has the smaller propagation factor for all three molecules. Relative to $KL+X_\mu$, it reduces $\hat{F}_\Sigma$ by factors of 1.34, 2.70, and 2.15 for NMA, glycerol, and neutral diethanolamine, respectively, and reaches $t = 1$ in fewer accepted stages in every case. The corresponding

molecule (d)	start ρ_0	end ρ_1	KL+X $_{\mu}$			KL+X $_{\mu}$ +X $_{(\hat{\mu}+\hat{\nu})/2}$		
			$\hat{F}_{\Sigma}$	stages	time	$\hat{F}_{\Sigma}$	stages	time
NMA (30)	(50, 0.20)	(100, 0.15)	20.19	6	6.0	15.12	5	12.1
glycerol (36)	(50, 0.20)	(100, 0.10)	146.2	9	16.5	54.16	8	35.8
neutral diethanolamine (48)	(50, 0.20)	(100, 0.10)	303.1	12	23.1	140.9	9	43.1

Table 7: Boltzmann generator training results for the additional achiral molecules, including propagation factor $\hat{F}_{\Sigma}$, the number of accepted stages, and total wall time in minutes. The two method groups use the full losses KL+X $_{\mu}$ and KL+X $_{\mu}$ +X $_{(\hat{\mu}+\hat{\nu})/2}$, respectively. Each row reports one seed and one sharpening path; bold marks the smaller $\hat{F}_{\Sigma}$ for that molecule.

KLXX runs require 2.02, 2.16, and 1.86 times the recorded wall time. Thus the factor reduction is accompanied by the additional QT and optimization work in KLXX.

Figure 10 resolves each factor into its accepted-stage contributions. Because the two losses use method-specific adaptive stage schedules, the individual arcs are descriptive rather than transition-matched comparisons. Sharpening can be the tighter operation at intermediate stages, so the total factor includes both ESS values reported on each arc.

The final regularized samples are compared with OpenMM PT references, taken at 300 K, in Figure 11. The dashed blue curves are plotted above the solid red curves so that close agreement between samples from the two losses remains visible. The selected torsions resolve the NMA amide trans basin and the trans/gauche wells of glycerol and neutral diethanolamine.

The curves overlap closely on these selected one-dimensional marginals, but this diagnostic does not determine the complete molecular distributions. In particular, it does not resolve the NMA cis/trans weights, joint glycerol or neutral diethanolamine rotamers, or unrestricted signed-volume statistics. The reference frames are also correlated, and every generator configuration has one training seed.

6.5 Chiral molecule results

Molecular setup. We report completed runs for alanine dipeptide in its L form, (2*R*, 3*R*)-2,3-butanediol, and Ac-Pro-NHMe. Only the KLXX loss is evaluated, so this subsection contains no within-target loss comparison. Every run uses seed 0, 250 Adam steps per accepted flow stage, an 8-level SMC ladder, and 100 mixed-domain MALA steps. The validation set and batch size pairs are $3.0 \times 10^5 / 1.5 \times 10^4$, $2.4 \times 10^5 / 1.2 \times 10^4$, and $3.2 \times 10^5 / 1.6 \times 10^4$, respectively. After training, each stage flow is replayed on a larger particle set, giving 10^7 inference samples for alanine dipeptide and 2,3-butanediol and 1.5×10^7 for Ac-Pro-NHMe. The surfaces below use those inference sets, on which the endpoint-to-physical weights are nearly uniform, with RESS equal to 1.000000, 0.999981, and 1.000000, respectively, so the regularized endpoint and the physical potential are indistinguishable on the sampled region.

The training and replay costs are deliberately asymmetric. Training never carries more than 3.2×10^5 samples, which keeps the wall times of Table 8 modest, while the inference sets are produced afterwards by replaying frozen flows without backpropagation. Replay is nonetheless the more expensive half in absolute terms, because sampling evaluates the NSF inverse G^{-1} , the expensive direction for this flow structure. Cost therefore sits where it is easiest to pay: replay is a fixed number of independent map evaluations and parallelizes across devices, whereas the training schedule is sequential in the stage index and the reference simulations of Section 6.3 must advance their trajectories sequentially in time.

All three runs reach $t = 1$; Table 8 gives their sharpening paths and stage summaries. The propagation factor is accumulated over the flow and sharpening ESS of every accepted stage, as in the earlier molecular tables. The three rows are different molecules with different validation sizes and regularization paths, so they are not comparable with one another.

Figure 12 shows one representative final-stage conformation per target, with gold rings on the fixed stereocenters: the *S* alanine C_{α} , both *R* centers of 2,3-butanediol, and the *S* proline C_{α} .

All three chiral targets carry an independent equilibrium reference; the first is published and the other two are our own. For alanine dipeptide it is the implicit-solvent trajectory data released with the FAB

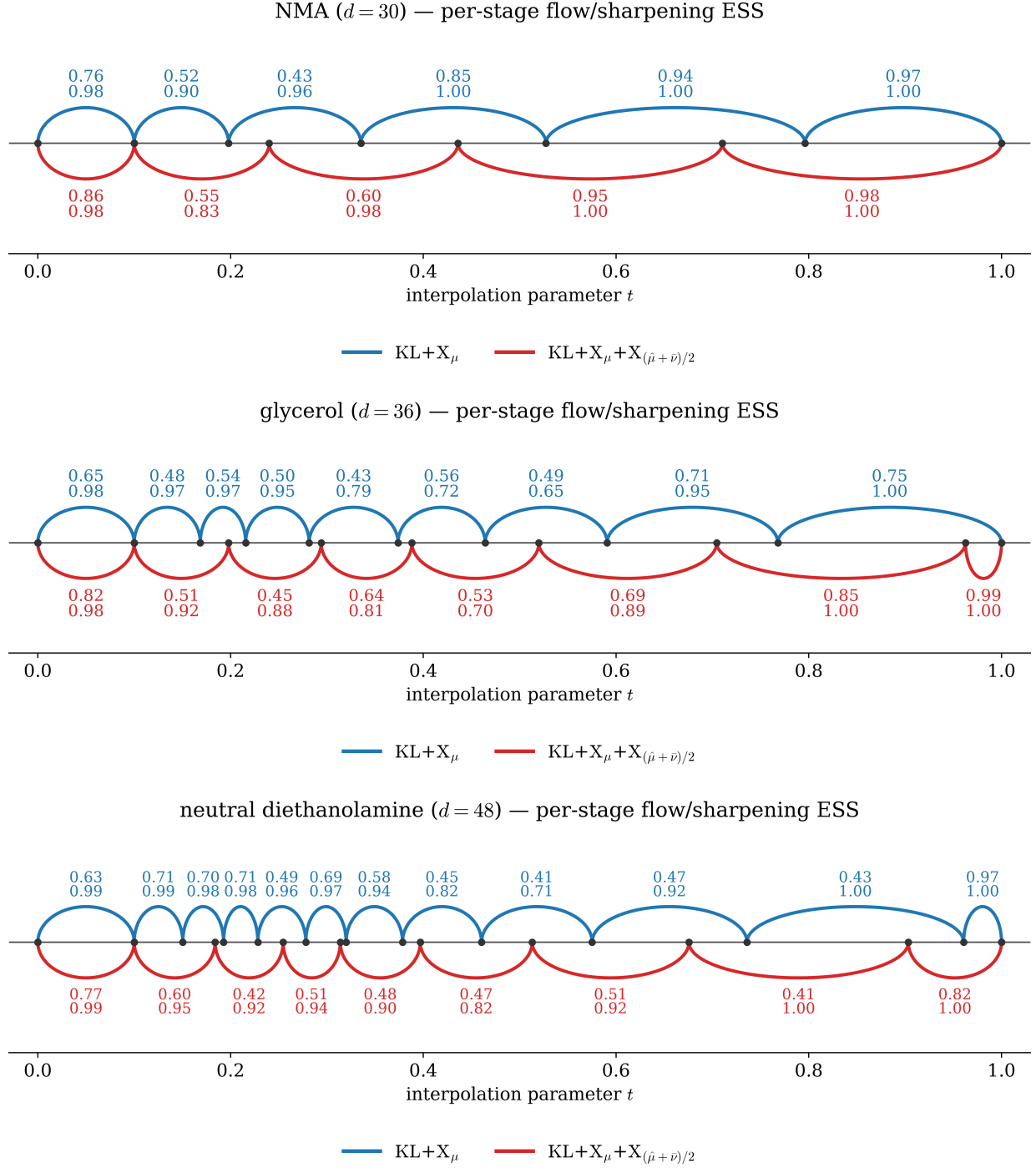


Figure 10: Per-stage flow and sharpening ESS for NMA (top), glycerol (middle), and neutral diethanolamine (bottom). Each arc spans one accepted transition $t_{k-1} \rightarrow t_k$. Within each stacked annotation, the first row is flow ESS and the second row is sharpening ESS. Blue upper arcs show $\text{KL} + \text{X}_\mu$; red lower arcs show $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \hat{\nu})/2}$. The horizontal coordinate is the source-to-target interpolation parameter $t \in [0, 1]$.

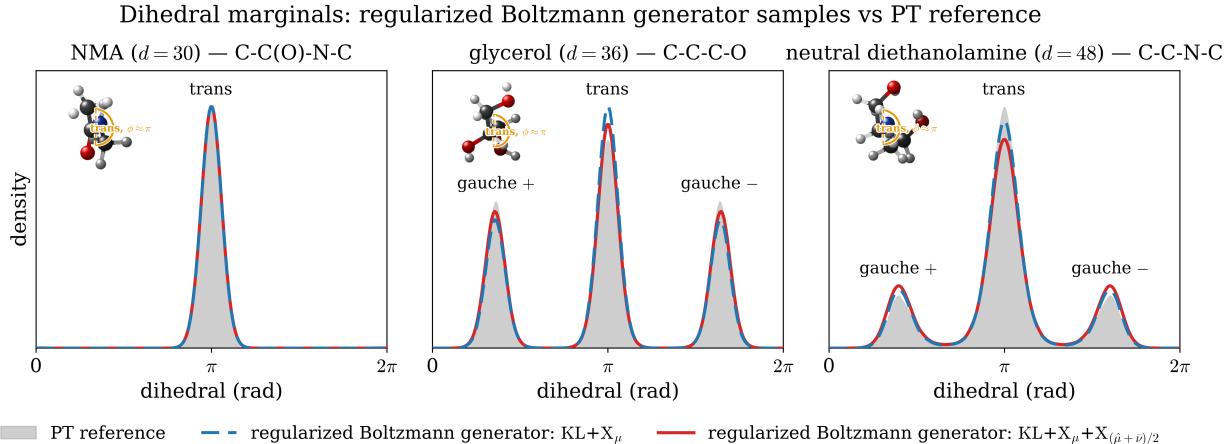


Figure 11: Selected dihedral marginals from regularized Boltzmann generator samples and OpenMM PT references for NMA, glycerol, and neutral diethanolamine. Gray denotes the reference; dashed blue denotes $KL+X_\mu$, and solid red denotes $KL+X_\mu+X_{(\hat{\mu}+\hat{\nu})/2}$. The upper-left insets show representative trans conformers. The NMA reference is a PT trajectory initialized in the trans basin, whereas the glycerol and neutral diethanolamine references pool two seeds. The NMA panel is therefore a trans-basin comparison rather than an estimate of the equilibrium cis/trans populations.

molecule (d)	start ρ_0	end ρ_1	$KL+X_\mu+X_{(\hat{\mu}+\hat{\nu})/2}$		
			$\hat{F}_\Sigma$	stages	time (min)
alanine dipeptide (60)	(50, 0.25)	(125, 0.10)	68.08	10	72.7
(2 <i>R</i> , 3 <i>R</i>)-2,3-butanediol (42)	(50, 0.20)	(100, 0.10)	37.45	8	35.8
Ac-Pro-NHMe (72)	(50, 0.25)	(150, 0.10)	176.7	11	124.7

Table 8: Boltzmann generator results for the chiral molecules, including propagation factor $\hat{F}_\Sigma$, the number of accepted stages, and total stage time in minutes. The method uses the loss $KL+X_\mu+X_{(\hat{\mu}+\hat{\nu})/2}$. Each row reports one seed and one sharpening path.

study (Midgley et al., 2023), generated for the same Hamiltonian and used here as ground truth, with 10^7 samples. For 2,3-butanediol it is a PT run on the unregularized physical potential at 300 K, six temperatures on a geometric 300–800 K grid, retaining 10^5 frames over 50.0 ns after a discarded 0.5 ns equilibration, with mean per-pair exchange probability 0.529 and 2.2×10^4 transitions among the three central rotamers. For Ac-Pro-NHMe the ACE-PRO amide barrier is about $26 k_B T$, so neither a plain trajectory nor PT crosses it. Its reference instead adds to the 300 K Langevin dynamics a Metropolis rigid π rotation of the proline side of the amide bond; that move is an involution with unit Jacobian, so the jump is exact and the target is never deformed. It retains 2×10^6 samples over 501.0 ns at 0.2028 jump acceptance.

Results comparison. Figure 13 tracks the alanine dipeptide Ramachandran surface along four accepted sharpening stages. Table 9 compares the final KLXX stage against ground truth on the same 100×100 grid over (ϕ, ψ) with the same smoothing, each surface normalized to its own most populated bin, so only per-method values are quoted. The three populations are raw bin counts on that grid, taken before any smoothing: α_L is the half-plane $\phi > 0$, and the bins with $\phi \leq 0$ are split by the sign of ψ into β /PPII and α_R . The $\phi > 0$ half-plane is the sparse region of this target. It holds 0.33% of the ground-truth mass, a factor of about 240 below β /PPII, and the KLXX Boltzmann generator places 0.0031 of its own mass there against the ground-truth 0.0033, so a region of this weight is reproduced at its own scale rather than left empty or overfilled. The three populations differ by at most 0.0018, and the mean free energies agree to $0.03 k_B T$ in every band. A single target does not by itself establish accuracy across systems.

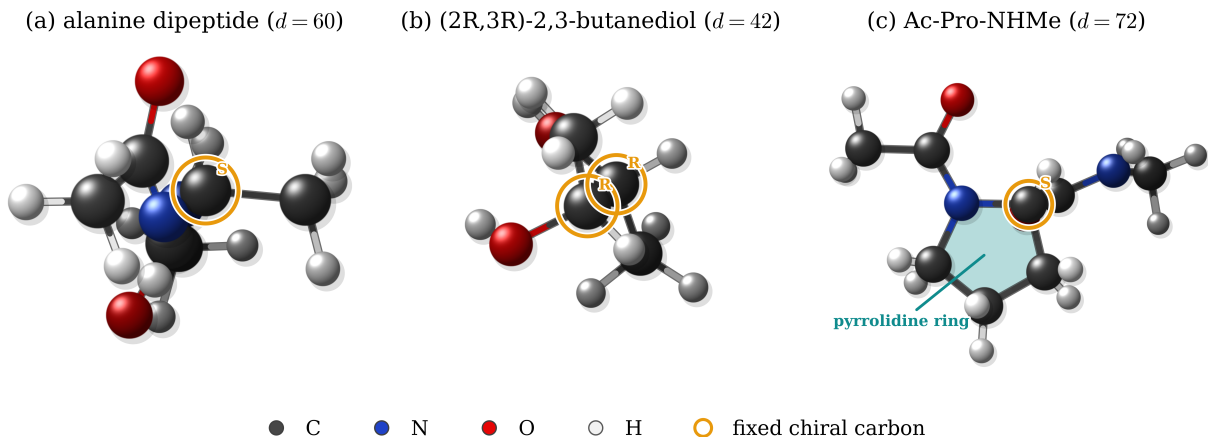


Figure 12: Representative final-stage conformations for the three completed chiral targets. From left to right: L-alanine dipeptide, (2*R*,3*R*)-2,3-butanediol, and Ac-Pro-NHMe. Gold rings and the adjacent *R/S* labels identify every fixed stereocenter; the atom colors follow the CPK convention shown in the legend. Each structure is the validation sample nearest the modal center of a molecule-specific two-dihedral histogram.

sample set	backbone basin population		
	α_L	β/PII	α_R
KL+ $X_\mu + X_{(\bar{\mu}+\bar{\nu})/2}$	0.0031	0.7958	0.2012
ground truth	0.0033	0.7940	0.2027
sample set	mean free energy over reference-classified bins		
	low	high	all
KL+ $X_\mu + X_{(\bar{\mu}+\bar{\nu})/2}$	1.230	6.323	5.379
ground truth	1.230	6.293	5.355

Table 9: Alanine dipeptide at the final accepted stage against ground truth, the implicit-solvent reference data of Midgley et al. (2023), both with 10^7 samples. Basins split the (ϕ, ψ) plane by $\phi > 0$ for α_L and, for $\phi \leq 0$, by the sign of ψ . Bins are classified by the reference surface as low below $2k_B T$ and high above $3k_B T$, and restricted to bins finite in both surfaces; entries are mean free energies in units of $k_B T$.

Figure 14 compares the accepted 2,3-butanediol stage nearest $t = 0.5$ with the final stage; both rows resolve the heavy-atom rotamers and hydroxyl orientations. Table 10 compares the final KLXX stage against the 2,3-butanediol reference on the same 100×100 torsion grid with the same smoothing. The two sample sets differ in size by two orders of magnitude, and the two halves of the table are read accordingly. A basin population and a mean over occupied bins are ratios, estimated without bias at either size, so both are quoted from the full 10^7 inference set. A maximum free energy is not such a quantity: each surface is normalized to its own most populated bin, so the deepest value it can resolve is set by the bin holding a single sample. Maxima are quoted only at matched count, with the same inference set thinned to the reference’s 10^5 .

KLXX and the 2,3-butanediol reference agree closely: the rotamer populations differ by at most 0.020, in the dominant trans well; the mean free energies by at most $0.23 k_B T$; and at matched sample count the maxima by $0.02 k_B T$ and $0.00 k_B T$. The residual trans difference is larger than the reference’s own sampling error, which the rotamer transition count places near 0.003, so it is not attributable to reference noise alone.

Ac-Pro-NHMe adds a ring degree of freedom and a cis/trans peptide torsion. Figure 15 compares the accepted stage nearest $t = 0.5$ with the final stage. Table 11 compares the final KLXX stage against the Ac-Pro-NHMe reference. The two sets agree on the qualitative picture and disagree quantitatively on the minority sector: both populate cis and trans with trans as the majority, and the two cis-to-trans free energy differences are within 0.53 kJ/mol of each other, while the cis populations differ by 0.0244.

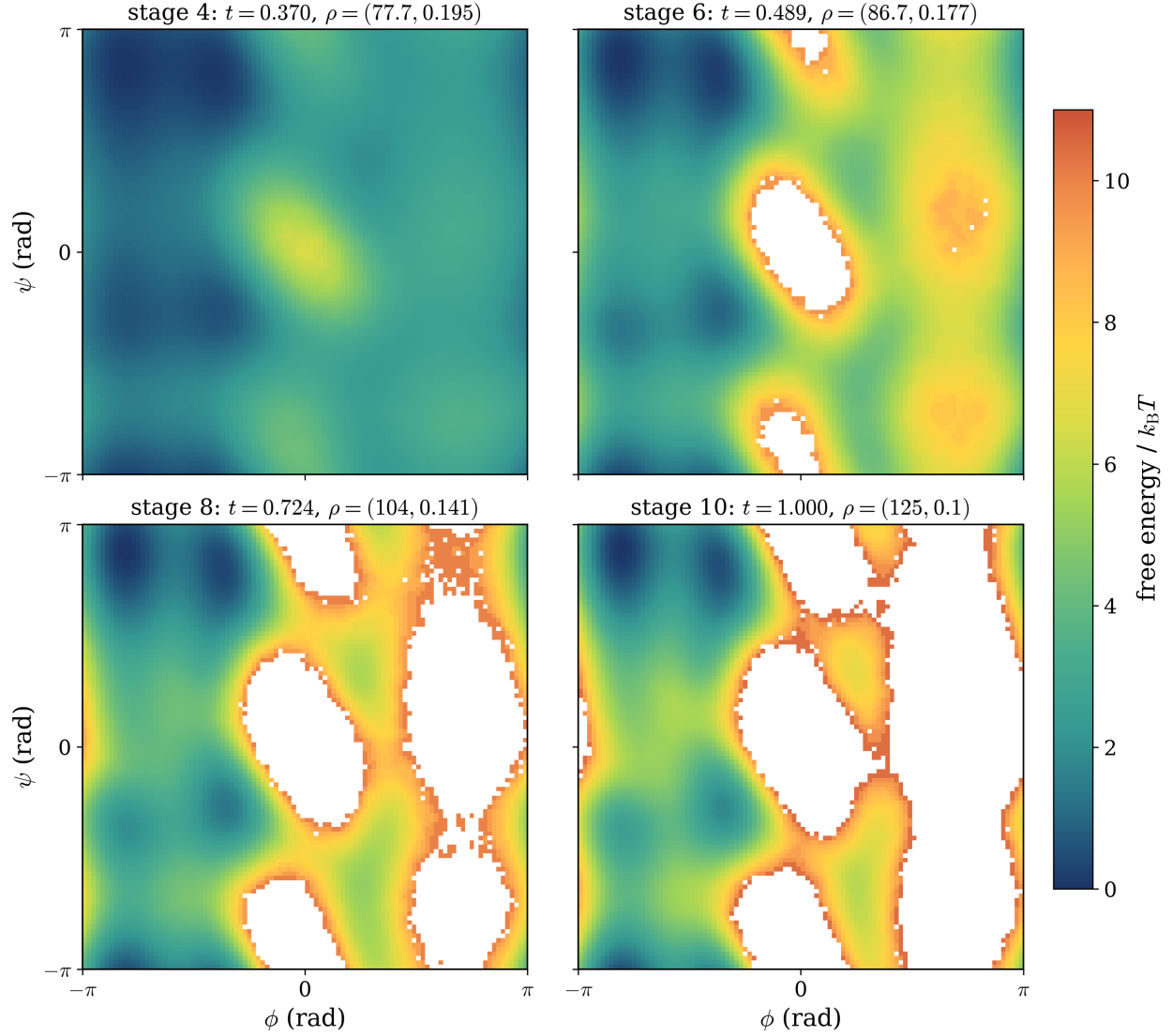


Figure 13: Alanine dipeptide Ramachandran free energy surfaces at accepted stages 4, 6, 8, and 10. Each panel contains 10^7 inference samples obtained by generating pushforward samples with the corresponding stage flow and then applying resampling, rejuvenation, sharpening, and post-sharpen rejuvenation. The panel titles give the interpolation parameter and regularization value; stage 10 reaches $t = 1$ and $\rho = (125, 0.10)$. A common colorbar reports free energy relative to the observed minimum in units of $k_B T$.

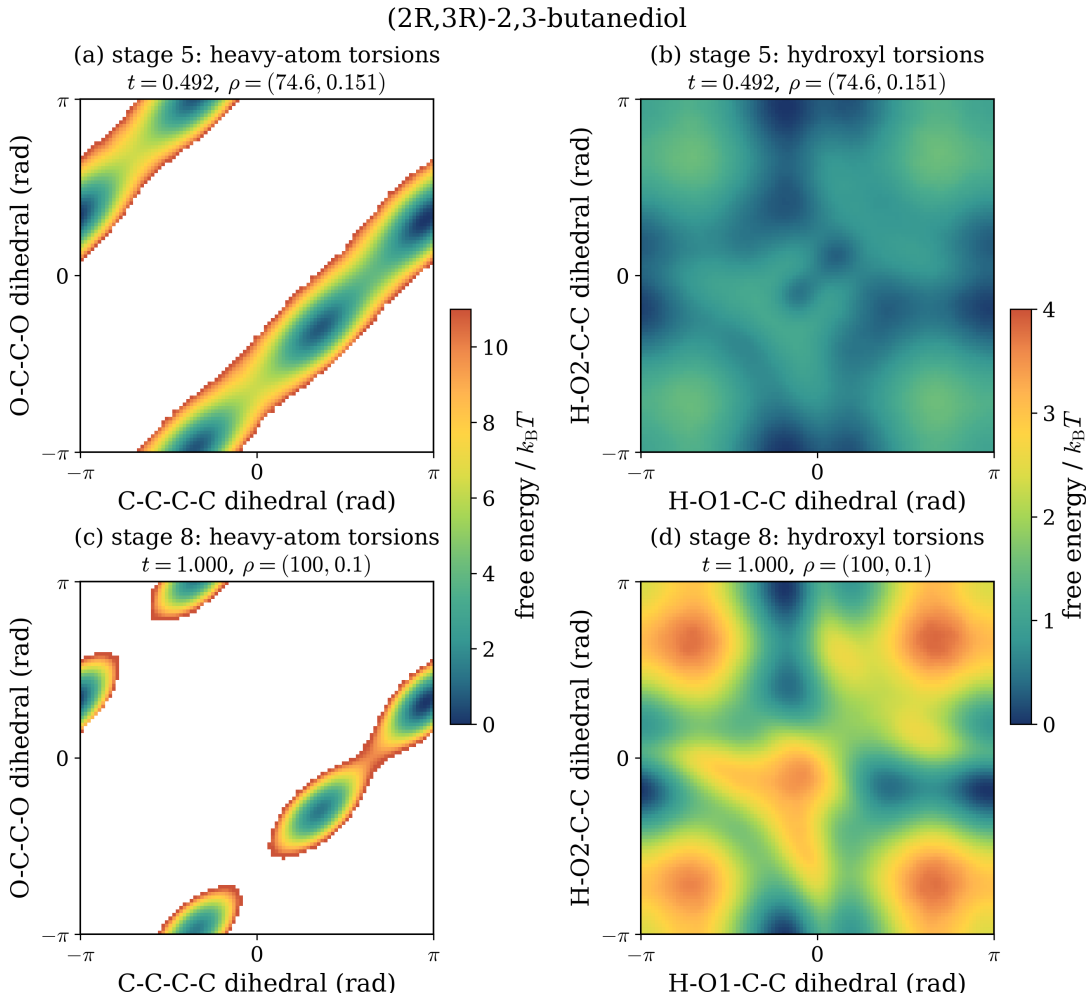


Figure 14: Conformational free energy surfaces for (2R,3R)-2,3-butanediol from 10^7 inference samples per stage. The upper row is stage 5, the accepted stage nearest $t = 0.5$, at $t = 0.492$ and $\rho = (74.6, 0.151)$; the lower row is the final stage 8 at $t = 1$ and $\rho = (100, 0.10)$. The left column shows the C-C-C-C and O-C-C-O heavy-atom torsions, and the right column shows the two hydroxyl-orientation torsions. Each column shares its scale across stages, and the two diagnostics have separate colorbars in units of $k_B T$.

These figures and tables diagnose the regularized KLXX sample sets and their retained stereochemical components. What this subsection does not provide is a matched loss comparison or uncertainty over independently trained generators.

Ac-Pro-NHMe residual gap. Ac-Pro-NHMe is the most difficult of the three chiral targets. Its two peptide sectors are separated by the ACE-PRO amide barrier of about $26 k_B T$, which no unbiased local move crosses, so even the reference needs a jump move built for that one bond before it can equilibrate them. The KLXX Boltzmann generator is given none of that: no reference frames, no bias potential, no elevated temperature, and no conformational prior, and it still populates both peptide sectors, keeps the S proline C_α , and orders the sectors correctly.

The residual gap is what one expects from a finite sample rather than from a systematic error in the method. By Theorem 2 the flow and resample inference is asymptotically unbiased: its error is bounded at rate $N^{-1/2}$ for every bounded observable and is removed by enlarging the sample alone, so a discrepancy that survives at fixed N is the finite-sample term and should shrink as N grows. Its size is not predicted by

sample set	central C-C-C-C rotamer population		
	gauche−	gauche+	trans
KL+X _μ +X _{(μ+ν)/2}	0.1486	0.1861	0.6652
OpenMM reference	0.1425	0.1726	0.6849
sample set	mean free energy over reference-classified bins		
	low	high	all
KL+X _μ +X _{(μ+ν)/2}	1.229	5.126	4.137
OpenMM reference	1.259	5.353	4.309

Table 10: (2*R*, 3*R*)-2,3-butanediol at the final accepted stage against an independent OpenMM PT reference, with 10^7 inference samples and 10^5 reference frames. Rotamers split the central C-C-C-C torsion at its eclipsed barriers, trans for $|\phi| \geq 2\pi/3$ and one gauche well on either side of $\phi = 0$. Bins are classified by the reference surface as low below $2k_B T$ and high above $3k_B T$, and restricted to bins finite in both surfaces; entries are mean free energies in units of $k_B T$ on the heavy-atom torsions.

sample set	cis	trans	$\Delta G(\text{cis} - \text{trans})$
KL+X _μ +X _{(μ+ν)/2}	0.1199	0.8801	4.97
OpenMM reference	0.1443	0.8557	4.44

Table 11: Ac-Pro-NHMe peptide torsion populations at the final accepted stage against an independent OpenMM reference, with 1.5×10^7 inference samples and 2×10^6 reference samples. Cis is $|\omega| < \pi/2$, counted per sample on both sides. Free energy differences are in kJ/mol.

the propagation factors of Table 8, where $\hat{F}_\Sigma$ is 176.7 for Ac-Pro-NHMe against 68.08 for alanine dipeptide, a ratio of about 2.6. Both comparisons are basin populations, bounded observables with $\text{osc}(\varphi) = 1$, so the observable is not what separates them; what does not carry over is the factor. As Section 4.4 and Appendix B.3 state, $\hat{F}_\Sigma$ is assembled from the per-stage ESS and is not an upper bound on the constant of (23), which depends on the stage weights through their suprema. The r^{-12} Lennard–Jones core of (31) is what makes large weights available here: the soft cap of (34) slows the core’s growth to logarithmic but leaves unbounded the potential difference that the stage weight exponentiates, so the quantity the constant depends on can be large while the reported factor stays small. A factor larger by 2.6 therefore does not predict a bias larger by 2.6, and the near-singular core of the potential explains the residual gap more plausibly than the dimension of the target.

Enlarging the inference set beyond 1.5×10^7 is the direct test, and it is the one we do not run. The replay cost grows in proportion to the sample count, this target is already the most expensive of the three, and pushing N far enough to resolve a difference of 0.02 in a population of 0.14 is beyond the budget of this study. We leave the gap open rather than close it by tuning.

7 Conclusions

Forward KL is mass-covering for exact distributions, but its sampled loss sees only the target regions supplied to it. This gap can produce a fake ESS, transfer surrogate error to the flow, and leave target-poor proposal regions weakly constrained. Log-ratio variation addresses these failures by comparing the exact log-density ratio at pairs of samples. The weighting measure determines which regions are compared: target samples measure density variation on reached target regions, QT samples add candidate modes, and pushforward samples expose regions occupied by the proposal. KLXX combines these comparisons while retaining the exact target as a minimizer and the flow as a tractable importance proposal.

The theory establishes a narrower statement. For the FR gradient flow of forward KL with X_μ , the regularizer adds a nonpositive term to the instantaneous KL dissipation. For a fixed smooth surrogate and $\lambda < \frac{1}{2}$, it also subtracts a nonnegative discrepancy term from the [stationary error](#) bound. This does not

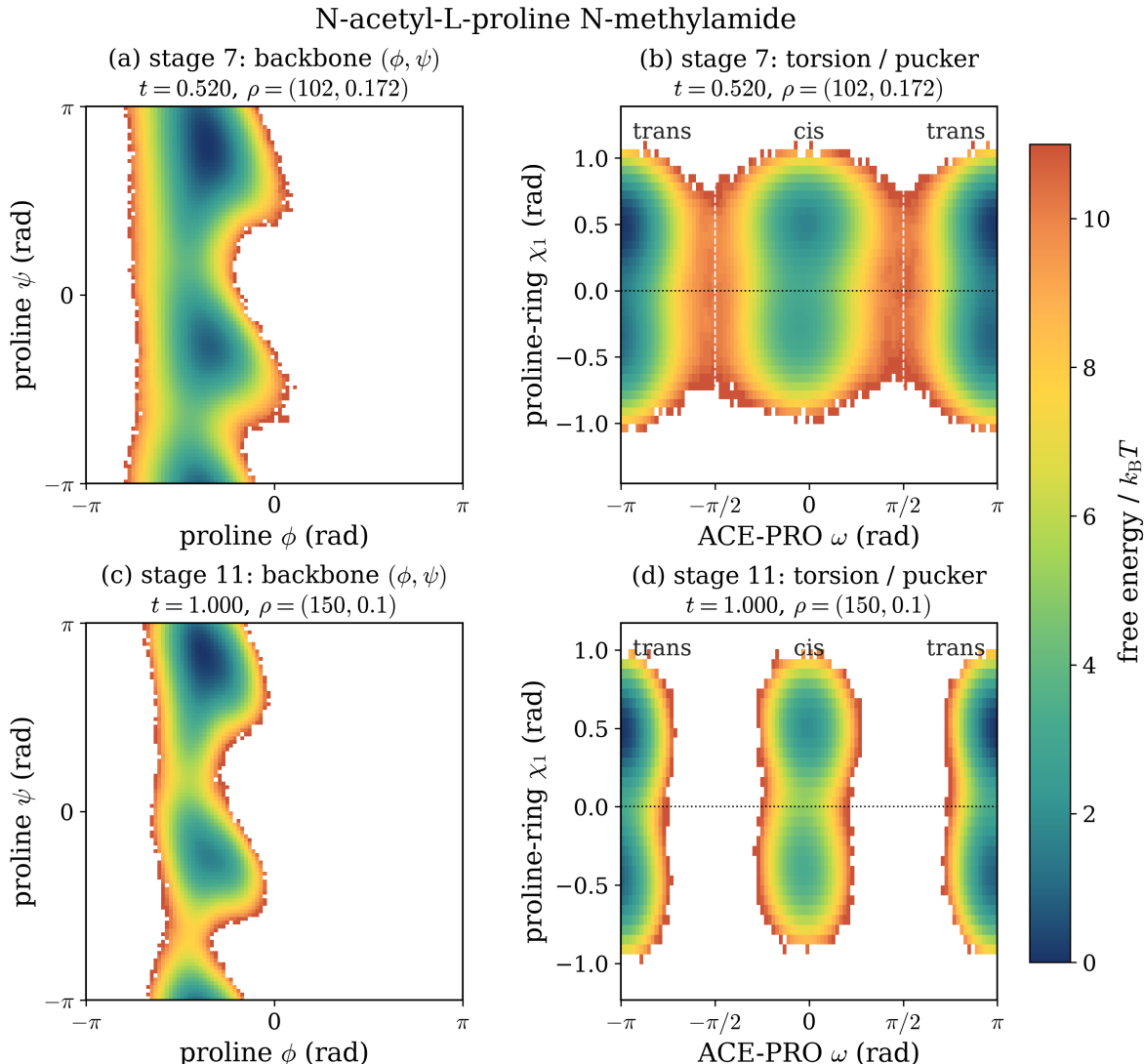


Figure 15: Conformational free energy surfaces for Ac-Pro-NHMe from 1.5×10^7 inference samples per stage. The upper row is stage 7, the accepted stage nearest $t = 0.5$, at $t = 0.520$ and $\rho = (102, 0.172)$; the lower row is the final stage 11 at $t = 1$ and $\rho = (150, 0.10)$. The left column shows the proline backbone (ϕ, ψ) surface, and the right column shows the joint ACE-PRO peptide torsion ω and proline-ring pucker χ_1 , with the cis and trans sectors marked. All four panels share one free energy scale in units of $k_B T$.

prove finite-step behavior for full KLXX, Adam, or the experimental value $\lambda = 1$. Theorem 2 is separate and complete: it bounds the L^2 error of the flow and resample scheme at rate $N^{-1/2}$ for every bounded observable, with a constant independent of N , so the scheme is asymptotically unbiased. The propagation factors read from that constant summarize stagewise weight degeneracy, and the reported $\hat{F}_\Sigma$ is a computable surrogate rather than a bound on the error constant.

The numerical results provide the remaining evidence. The 2D and ϕ^4 ablations are consistent with the roles of the target, QT, and pushforward samples; the product test shows higher validation ESS for every loss with log-ratio variation at all sixteen tested dimensions. On the shared clock stage schedules, KLXX has higher validation ESS on all 28 shared stages and a smaller propagation factor at every reported batch size. In the single-seed alkane runs, KLXX has the smallest factor in all ten matched settings, while forward KL approaches the identity baseline as molecule size grows. The sharpening paths have lower KLXX factors for the larger alkanes. For NMA, glycerol, and neutral diethanolamine, KLXX also has a

smaller factor than $\text{KL} + X_\mu$ and reaches the endpoint in fewer accepted stages. The three chiral KLXX runs reach $t = 1$, retain their specified stereochemical components, and populate multiple torsional basins in the reported conformational diagnostics. For alanine dipeptide, the final sample set matches the published equilibrium reference of Midgley et al. (2023) at 10^7 samples per side: the $\phi > 0$ half-plane, which holds 0.33% of the reference mass, is reproduced at 0.0031 against 0.0033, the three backbone basin populations differ by at most 0.0018, and the mean free energies agree to $0.03 k_B T$ in every band. For Ac-Pro-NHMe, the hardest target and the one reached with no external data and no conformational prior, the KLXX Boltzmann generator populates both peptide sectors and orders them correctly, and its cis population differs from an independent reference of our own by 0.0244. Selected alkane observables and achiral dihedral marginals agree with independent physical-potential simulations on the reported diagnostics. These findings are consistent with an explanation based on additional support information; they are not a theorem of universal dominance or a fixed-compute comparison.

The clock and molecular configurations require additional training seeds to measure run-to-run variation. The finite alkane endpoints have nearly uniform endpoint-to-physical weights on the reported sample sets, but a continuation to U would remove the remaining regularization. The additional achiral figures report selected one-dimensional marginals and do not determine joint conformational weights. Among the chiral results, alanine dipeptide is checked against published ground truth, and 2,3-butanediol and Ac-Pro-NHMe against references of our own. The Ac-Pro-NHMe comparison agrees on the qualitative picture and leaves a quantitative gap in the cis population; the propagation bound makes that gap a finite-sample term, but resolving it would require an inference set beyond the budget of this study, so we report it open. No alternative-loss run was made for any of the three. Larger molecules and explicit solvent remain outside the completed results.

Acknowledgments

The normalizing flow computations use the JAX backend through two Python packages maintained by Xuda Ye. The packages `jflows` and `jflows-md` can be installed directly from PyPI with:

```
pip install jflows jflows-md
```

Molecular benchmark data were generated with OpenMM and AmberTools, and the alanine dipeptide reference data were taken from the FAB study (Midgley et al., 2023). The source files are available at:

<https://github.com/xuda-ye-math/KLXX>

The GPU-accelerated runs used one MSI GeForce RTX 5090 32G Gaming Trio OC with 32 GB of VRAM. The card ran at a 600 W power limit with clock offsets of +250 MHz on the core and +2000 MHz on the memory. All reported wall times correspond to this overclocked configuration.

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A Fisher–Rao gradient flow with log-ratio variation

Section 2 states the main gradient flow results used in the paper. This appendix supplies their derivations and proofs on the space of densities. We view ν as a generic probability density evolving along the gradient flow of the loss under an appropriate Riemannian metric on the space of densities (Ambrosio et al., 2008; Villani, 2009) and ask how fast ν_t converges to the target μ in $\text{KL}(\mu\|\nu_t)$.

At ties we set $\text{sgn}(0) = 0$. Throughout this appendix, the densities are assumed positive and sufficiently regular for the displayed gradient calculations. We restrict to forward KL with the regularizer X_μ ,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (41)$$

where $\lambda > 0$ is fixed. Define the importance weight with respect to ν by

$$w(x) := \frac{\mu(x)}{\nu(x)},$$

so that the log-ratio is $z(x) = \log w(x) = \log \mu(x) - \log \nu(x)$, the field z of (6). The importance weight w is distinct from the weighting density ω in X_ω .

In the Wasserstein-2 geometry, induced by optimal transport (Jordan et al., 1998; Otto, 2001), the forward KL fails to be displacement convex along Wasserstein-2 geodesics, and the standard Otto–Villani/Bakry–Émery argument (Otto and Villani, 2000; Bakry et al., 2014) used for the reverse KL does not apply. We therefore work in the FR geometry (Amari, 2016; Bauer et al., 2016). Along the positive flow, the forward KL has an exponential bound and X_μ contributes a nonlocal birth–death term (Lu et al., 2019): the update at x depends on how its importance weight compares with the weights under μ .

The FR gradient of a functional $\mathcal{F}[\nu]$ and the associated dissipation identity along the FR gradient flow $\partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{F}(\nu_t)$ are given by

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left(\frac{\delta \mathcal{F}}{\delta \nu} - \mathbb{E}_\nu \left[\frac{\delta \mathcal{F}}{\delta \nu} \right] \right), \quad \frac{d}{dt} \mathcal{F}[\nu_t] = - \int_{\mathbb{R}^d} \left(\frac{\delta \mathcal{F}}{\delta \nu} - \mathbb{E}_{\nu_t} \left[\frac{\delta \mathcal{F}}{\delta \nu} \right] \right)^2 \nu_t dx. \quad (42)$$

The mean-subtraction enforces $\int \text{grad}^{\text{FR}} \mathcal{F} = 0$, so the FR gradient flow conserves probability mass.

A.1 Derivation of X_μ -regularized Fisher–Rao gradient flow

The first variation of the forward KL with respect to ν is

$$\frac{\delta \text{KL}(\mu\|\nu)}{\delta \nu}(x) = -\frac{\mu(x)}{\nu(x)} = -w(x), \quad (43)$$

so by (42), using $\mathbb{E}_\nu[w] = \int_{\mathbb{R}^d} \mu dx = 1$, the FR gradient of the forward KL is

$$\text{grad}^{\text{FR}} [\text{KL}(\mu\|\nu)](x) = \nu(x) - \mu(x). \quad (44)$$

For X_μ , the variation uses $\text{sgn}(0) = 0$ at ties. At an evaluation point r , the variation of the inner log-ratio difference is

$$\frac{\delta}{\delta \nu(r)} [\log w(x) - \log w(y)] = -\frac{\delta(r-x)}{\nu(x)} + \frac{\delta(r-y)}{\nu(y)}.$$

Here δ is the Dirac distribution, so integration against $\delta(r-x)$ evaluates the remaining integrand at $x=r$. Applying this sign convention and then exchanging x and y in the second term gives

$$\begin{aligned} \frac{\delta X_\mu(\mu\|\nu)}{\delta \nu}(r) &= \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) \text{sgn}(w(x) - w(y)) \left[-\frac{\delta(r-x)}{\nu(x)} + \frac{\delta(r-y)}{\nu(y)} \right] dx dy \\ &= -2w(r) \int_{\mathbb{R}^d} \mu(y) \text{sgn}(w(r) - w(y)) dy. \end{aligned}$$

Hence the variation is

$$\frac{\delta X_\mu(\mu\|\nu)}{\delta\nu}(x) = -2w(x)S(x), \quad S(x) := \mathbb{E}_{y\sim\mu}[\text{sgn}(w(x) - w(y))]. \quad (45)$$

We call S the *nonlocal correlation* of the importance weight under μ : $S(x) > 0$ indicates that $w(x)$ is comparatively large within the distribution of weights, and hence that ν pointwise undercovers μ relative to that distribution; $S(x) < 0$ has the opposite interpretation, and $S \equiv 0$ at $\nu = \mu$. By antisymmetry,

$$\mathbb{E}_\mu[S] = \mathbb{E}_{x,y\sim\mu}[\text{sgn}(w(x) - w(y))] = 0,$$

and the FR gradient of X_μ with respect to ν is

$$\text{grad}^{\text{FR}}[X_\mu(\mu\|\nu)](x) = -2\mu(x)S(x). \quad (46)$$

The FR gradient of $\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu)$ is the sum of (44) and (46):

$$\text{grad}^{\text{FR}}[\mathcal{L}](x) = \nu(x) - \mu(x) - 2\lambda\mu(x)S(x).$$

This yields the FR gradient flow (11) stated in Section 2. It is a nonlocal partial differential equation: its right-hand side at x depends on $S_t(x)$, which requires the entire profile of the importance weight μ/ν_t over the support of μ . Mass is conserved along the flow: integrating (11) and using $\int_{\mathbb{R}^d} \mu(x)S_t(x)dx = 0$ gives

$$\frac{d}{dt} \int_{\mathbb{R}^d} \nu_t(x)dx = 1 - \int_{\mathbb{R}^d} \nu_t(x)dx,$$

so a unit mass at $t = 0$ is preserved for all t , and the flow is stationary at $\nu_t = \mu$.

A.2 Convergence analysis of the X_μ -regularized flow

To analyze the convergence rate of the regularized flow (11), we take $\text{KL}(\mu\|\nu_t)$ as the error metric and differentiate it along the flow:

$$\frac{d}{dt} \text{KL}(\mu\|\nu_t) = - \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} (1 + 2\lambda S_t) dx + \int_{\mathbb{R}^d} \mu dx = -\chi^2(\mu\|\nu_t) - 2\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx, \quad (47)$$

where the chi-squared divergence is

$$\chi^2(\mu\|\nu) := \int_{\mathbb{R}^d} (w - 1)^2 \nu dx = \int_{\mathbb{R}^d} w^2 \nu dx - 1.$$

The sign of the second integral in the dissipation (47) is settled by the following correlation identity.

Lemma 1 *Let μ and ν be positive probability densities with $\chi^2(\mu\|\nu) < \infty$. Then*

$$\int_{\mathbb{R}^d} \frac{\mu(x)^2}{\nu(x)} S(x) dx = \frac{1}{2} \mathbb{E}_{x,y\sim\mu}[|w(x) - w(y)|] \geq 0,$$

with equality if and only if w is μ -almost-everywhere constant.

Proof. Since $\mu^2/\nu = \mu w$, the definition of S gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \mathbb{E}_{x\sim\mu}[w(x)S(x)] = \mathbb{E}_{x,y\sim\mu}[w(x) \text{sgn}(w(x) - w(y))].$$

Symmetrizing in $x \leftrightarrow y$ gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \frac{1}{2} \mathbb{E}_{x,y\sim\mu}[(w(x) - w(y)) \text{sgn}(w(x) - w(y))] = \frac{1}{2} \mathbb{E}_{x,y\sim\mu}[|w(x) - w(y)|].$$

□

Applying Lemma 1 with $\nu = \nu_t$ and $S = S_t$, the dissipation (47) can be rewritten as

$$\frac{d}{dt} \text{KL}(\mu \| \nu_t) = -\chi^2(\mu \| \nu_t) - \lambda \mathbb{E}_{x, y \sim \mu} \left[\left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right]. \quad (48)$$

The elementary inequality $w \log w \leq w^2 - w$ applied to $w = \mu/\nu_t$ gives $\chi^2(\mu \| \nu_t) \geq \text{KL}(\mu \| \nu_t)$, and the second term of (48) is nonpositive by Lemma 1, so discarding it leaves $\frac{d}{dt} \text{KL}(\mu \| \nu_t) \leq -\text{KL}(\mu \| \nu_t)$, and Grönwall's lemma yields the exponential decay

$$\text{KL}(\mu \| \nu_t) \leq e^{-t} \text{KL}(\mu \| \nu_0), \quad t \geq 0, \quad (49)$$

with no structural assumption on μ : neither log-concavity, nor a spectral gap, nor any growth condition. This contrasts with the Wasserstein-2 geometry, where the forward KL is not displacement convex and an analogous exponential rate is unavailable without further assumptions.

However, the uniform convergence rate (49) should *not* be read as a blanket statement that minimizing the forward KL is intrinsically faster or more efficient than minimizing the reverse KL. The FR gradient flow is a continuous-time idealization on the space of densities and does not account for sampling cost. In practice, evaluating the forward KL and its gradient requires samples from μ , and producing representative samples in high dimensions is expensive. The bound above quantifies convergence given exact target expectations; it does not establish the behavior of this additional empirical term or remove the cost of constructing samples.

A naive comparison of dissipation rates between forward KL and $\text{KL} + \lambda X_\mu$ would also be misleading, since scaling the loss by a constant is mere time reparametrization with no qualitative gain. The substantive contribution of X_μ is structural rather than rate-based: the term from forward KL depends locally on the value of $\mu(x)/\nu_t(x)$, whereas the correlation S_t compares that value with the complete distribution of importance weights under μ . The regularized flow (11) therefore couples the update at x to this correlation, which the term from forward KL does not contain.

Two further cautions concern the discrete realization of the flow. First, the time-reparametrization argument cuts in both directions once the optimizer enters. Training follows Adam steps, not the FR gradient flow: the per-coordinate second-moment normalization absorbs a global rescaling of the loss, so a larger gradient buys no speedup by itself, while a regularizer that reshapes the relative magnitudes of the gradient components does change the optimizer's effective time scale. Part of any observed difference in training speed between forward KL and $\text{KL} + \lambda X_\mu$ may therefore be optimizer scaling rather than information geometry, and we claim no universal transient speedup on this basis. Second, the dissipation (48) is a density-space statement, while each gradient step uses a finite, generally biased sample estimate. The staged clock experiment examines this finite-batch setting with a schedule-matched comparison: its quantitative claims use the selected-proposal ESS over the validation set at each stage, not a composed endpoint estimate or a diagnostic computed on a training mini-batch.

Despite these cautions, the density-space flow retains a further monotonicity property: X_μ decreases along the regularized flow. Pairing the variation $\delta X_\mu / \delta \nu = -2wS$ (45) against the flow (11) gives

$$\frac{d}{dt} X_\mu(\mu \| \nu_t) = -2 \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx - 4\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t^2 dx \leq 0.$$

The first term is nonpositive by Lemma 1 and vanishes only when μ/ν_t is μ -almost-everywhere constant. Hence X_μ is nonincreasing along the flow, and strictly decreasing away from that case.

A.3 Asymptotic accuracy for a fixed biased surrogate

The convergence analysis of Section A.2 takes the outer expectation in $\text{KL}(\mu \| \nu) + \lambda X_\mu(\mu \| \nu)$ under the exact target. To isolate surrogate bias, fix a smooth density $\tilde{\mu}$ that does not vary with ν and define

$$\tilde{\mathcal{L}}[\nu] = \int_{\mathbb{R}^d} \tilde{\mu}(x) \log w(x) dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) |\log w(x) - \log w(y)| dx dy, \quad (50)$$

where the pointwise ratio $w(x) = \mu(x)/\nu(x)$ is still evaluated against the exact target. Repeating the gradient calculation of Section A.1 with μ replaced by $\tilde{\mu}$ only in the outer integrals, and using $\int_{\mathbb{R}^d} \tilde{\mu} \tilde{S} dx = 0$, which holds for every ν by antisymmetry of the sign, gives

$$\text{grad}^{\text{FR}}[\tilde{\mathcal{L}}](x) = \nu(x) - \tilde{\mu}(x)(1 + 2\lambda\tilde{S}(x)), \quad \tilde{S}(x) := \mathbb{E}_{y \sim \tilde{\mu}} [\text{sgn}(w(x) - w(y))].$$

A stationary density ν^* of this gradient flow therefore satisfies

$$\nu^*(x) = \tilde{\mu}(x) \left[1 + 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(y) \text{sgn}\left(\frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)}\right) dy \right]. \quad (51)$$

At $\lambda = 0$, the equation reduces to $\nu^* = \tilde{\mu}$ and the residual error is $\text{KL}(\mu \parallel \tilde{\mu})$. For $\lambda > 0$, the correlation $\tilde{S}^*$, which is $\tilde{S}$ evaluated at $\nu = \nu^*$, enters through $1 + 2\lambda\tilde{S}^*$, raising ν^* relative to $\tilde{\mu}$ where μ/ν^* is comparatively high and lowering it where the ratio is low. For $\lambda < \frac{1}{2}$, (51) is positive because $\tilde{S}^* \in [-1, 1]$. At the stationary density the antisymmetry above reads

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \tilde{S}^*(x) dx = \mathbb{E}_{x, y \sim \tilde{\mu}} \left[\text{sgn}\left(\frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)}\right) \right] = 0,$$

hence every stationary point $\nu^* = \tilde{\mu}(1 + 2\lambda\tilde{S}^*)$ is a probability density.

We now prove Theorem 1 of Section 2, whose fixed-surrogate loss (14) is the loss (50) derived above.

Proof. Substituting the stationary identity $\nu^* = \tilde{\mu}(1 + 2\lambda\tilde{S}^*)$ into the difference of forward KLs and using $\mu = \nu^* \cdot (\mu/\nu^*) = \tilde{\mu}(1 + 2\lambda\tilde{S}^*)(\mu/\nu^*)$,

$$\text{KL}(\mu \parallel \tilde{\mu}) - \text{KL}(\mu \parallel \nu^*) = \int_{\mathbb{R}^d} \mu(x) \log(1 + 2\lambda\tilde{S}^*(x)) dx = \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \varrho(x) \log \varrho(x) dx,$$

where $\varrho(x) := 1 + 2\lambda\tilde{S}^*(x) \geq 1 - 2\lambda > 0$, since $\tilde{S}^* \in [-1, 1]$ (a $\tilde{\mu}$ -average of signs) and $\lambda < \frac{1}{2}$. The elementary inequality $\varrho \log \varrho \geq \varrho - 1$ for $\varrho > 0$ then yields

$$\text{KL}(\mu \parallel \tilde{\mu}) - \text{KL}(\mu \parallel \nu^*) \geq 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx.$$

The symmetrization proving Lemma 1 uses only that the outer measure is a product of two copies of one density, so it applies here with outer density $\tilde{\mu}$ and ratio μ/ν^* , giving

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx = \frac{1}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy,$$

which gives (15). $\square$

At $\lambda = 0$, the bound is attained by $\nu^* = \tilde{\mu}$. For $0 < \lambda < \frac{1}{2}$, the subtracted discrepancy is nonnegative and is positive unless μ/ν^* is constant on the support of $\tilde{\mu}$. Thus, within the theorem's assumptions, X_μ improves the stationary error bound for a fixed biased surrogate.

The threshold $\lambda < \frac{1}{2}$ is a sufficient condition for strict positivity of ν^* . The experiments use $\lambda = 1$ and do not extend the theorem beyond its stated range. The live surrogate also changes with the current flow and is represented by finite batches, so it is not the fixed smooth density assumed here.

B Proof of the propagation factor analysis

Section 4.4 states the error bound and the two propagation factors read from it. The scheme analyzed throughout is flow and resample only: each accepted stage propagates every particle by the inverse stage map, reweights it, and resamples, and the MALA rejuvenation that the full generator applies after each resampling is omitted. No Markov kernel therefore acts between stages, no particle position is ever moved except by a stage map, and the quantity bounded is the propagation error alone.

This appendix simplifies that scheme to a resample-only procedure, restates the bound for it as Theorem 3, and proves that theorem; Theorem 2 follows at once. Section B.3 examines the factor the experiments report.

B.1 Notation and problem reduction

We write $\|\cdot\|_2$ for the L^2 norm over all the randomness of the algorithm, namely the N initial draws and every resampling draw. Without rejuvenation, resampling only copies particles and creates no new positions, and propagation applies G_k^{-1} to each of them. Every position present at stage k is therefore an initial draw with the inverse stage maps applied in order, so we write

$$\Theta_k := G_k^{-1} \circ \dots \circ G_1^{-1}, \quad \Theta_0 := \text{id}, \quad (52)$$

for that composition, so that every X_k^i is Θ_k applied to one of the initial draws.

Instead of studying the varying positions X_k^i , we push the stage- k target μ_k back through Θ_k . Define the distribution η_k and the density r_k by

$$\eta_k := (\Theta_k^{-1})_{\#} \mu_k, \quad r_k := w_k \circ \Theta_k, \quad (53)$$

for $k = 0, 1, \dots, K$, the density r_k for $k \geq 1$ only, so that $\eta_0 = \mu_0$, $(\Theta_k)_{\#} \eta_{k-1} = \nu_k$ and $r_k = d\eta_k/d\eta_{k-1}$. The scheme then becomes a resample-only procedure: η_k is obtained from η_{k-1} by multiplying by the importance weight r_k , no propagation step remains, and every stage acts on one fixed space. Nothing is lost, since

$$\|r_k\|_{\infty} = \|w_k\|_{\infty}, \quad \text{osc}(\varphi \circ \Theta_K) = \text{osc}(\varphi),$$

so neither the constant of Theorem 2 nor the observable is affected. The triple (μ_k, ν_k, w_k) is thus reduced to the pair (η_k, r_k) , with every flow step hidden.

Written out, the reduced procedure is the scheme of Section 4.4 with the propagation step deleted. Draw $Z_0^1, \dots, Z_0^N$ independently from $\eta_0 = \mu_0$ and, for $k = 1, \dots, K$, repeat two steps.

- *Reweighting.* Evaluate the reduced weight at each particle, $R_k^i = r_k(Z_{k-1}^i)$.
- *Resampling.* Draw $Z_k^{1:N}$ by multinomial resampling from $\sum_{i=1}^N (R_k^i / \sum_{l=1}^N R_k^l) \delta_{Z_{k-1}^i}$.

After stage K it returns the empirical measure $\eta_K^N = \frac{1}{N} \sum_{i=1}^N \delta_{Z_K^i}$ as the estimate of η_K , and we write $\eta_k^N = \frac{1}{N} \sum_{i=1}^N \delta_{Z_k^i}$ at every stage. Consequently, with $f := \varphi \circ \Theta_K$,

$$\mu_K^N(\varphi) = \eta_K^N(f), \quad \mu_K(\varphi) = \eta_K(f), \quad \text{osc}(f) = \text{osc}(\varphi), \quad (54)$$

so bounding the error of the reduced procedure bounds the error of the original. This is a change of variables and not a change of scheme: every resampling step is retained, with the same weights and the same draws.

Theorem 2 therefore becomes a statement about the reduced procedure, and we restate it as such.

Theorem 3 *Suppose $r_k > 0$ and the η_{k-1} -essential supremum $\|r_k\|_{\infty}$ is finite, for every accepted stage k . Then for every bounded observable f and every $N \geq 1$,*

$$\mathbb{E} \left[(\eta_K^N(f) - \eta_K(f))^2 \right] \leq \frac{C}{N}, \quad C = \frac{\text{osc}(f)^2}{4} \left(\sum_{k=0}^K \prod_{j=k+1}^K 3\|r_j\|_{\infty} \right)^2, \quad (55)$$

with the empty product equal to one.

The two statements are now the same one: (54) already gives $\mu_K^N(\varphi) = \eta_K^N(f)$ with $\text{osc}(f) = \text{osc}(\varphi)$, and $\|r_j\|_{\infty} = \|w_j\|_{\infty}$, so the two bounds in (23) and (55) have the same left-hand side and the same constant, and the hypotheses correspond in the same way. It is enough to prove Theorem 3.

One stage of the reduced procedure does two things to η_{k-1}^N : reweighting, which adds no randomness but distorts the measure, and resampling, which adds randomness but on average returns what it was given. The weighted distribution from which stage k resamples averages f to

$$\sum_{i=1}^N \frac{R_k^i}{\sum_{l=1}^N R_k^l} f(Z_{k-1}^i) = \frac{\eta_{k-1}^N(r_k f)}{\eta_{k-1}^N(r_k)}, \quad (56)$$

and the same ratio taken at η_{k-1} instead of η_{k-1}^N returns η_k , since $r_k = d\eta_k/d\eta_{k-1}$ and $\eta_{k-1}(r_k) = 1$. The denominator is thus one for the exact sequence but random for the particle one.

B.2 Proof of Theorem 3

The proof is an induction on the stage index. Two facts about the density r_k are used throughout. First, $\eta_{k-1}(r_k) = 1$, because $r_k = d\eta_k/d\eta_{k-1}$ and η_k is a probability measure. Second, the hypotheses of Theorem 3 give $\eta_{k-1}^N(r_k) \in (0, \|r_k\|_\infty]$, so no normalization vanishes or diverges.

The next lemma bounds the error of the weighted average (56) by the previous stage's unweighted errors on the fixed functions g_k and r_k , and so accounts for the error propagated by *reweighting*.

Lemma 2 *Let f be bounded and measurable, and define $g_k := r_k(f - \eta_k(f))$. Then $\eta_{k-1}(g_k) = 0$, $\text{osc}(g_k) \leq 2\|r_k\|_\infty \text{osc}(f)$, and*

$$\left| \frac{\eta_{k-1}^N(r_k f)}{\eta_{k-1}^N(r_k)} - \eta_k(f) \right| \leq |\eta_{k-1}^N(g_k) - \eta_{k-1}(g_k)| + \text{osc}(f) |\eta_{k-1}^N(r_k) - \eta_{k-1}(r_k)|. \quad (57)$$

Proof. Since $r_k = d\eta_k/d\eta_{k-1}$ we have $\eta_{k-1}(r_k h) = \eta_k(h)$ for bounded h . Using $\eta_{k-1}(r_k) = 1$, we obtain

$$\eta_{k-1}(g_k) = \eta_k(f) - \eta_k(f) \eta_{k-1}(r_k) = 0.$$

Write $D_k := \eta_{k-1}^N(r_k f)/\eta_{k-1}^N(r_k) - \eta_k(f)$ and $\delta_k := \eta_{k-1}^N(r_k) - 1$. Then

$$\eta_{k-1}^N(g_k) = \eta_{k-1}^N(r_k f) - \eta_k(f) \eta_{k-1}^N(r_k) = \eta_{k-1}^N(r_k) D_k = D_k(1 + \delta_k),$$

hence $D_k = \eta_{k-1}^N(g_k) - D_k \delta_k$, and the triangle inequality gives

$$|D_k| \leq |\eta_{k-1}^N(g_k)| + |D_k| |\delta_k|.$$

Both $\eta_{k-1}^N(r_k f)/\eta_{k-1}^N(r_k)$ and $\eta_k(f)$ are weighted averages of f and so lie in $[\inf f, \sup f]$, whence $|D_k| \leq \text{osc}(f)$. Using that bound on the right-hand side only, and keeping $|D_k|$ itself on the left, leaves

$$|D_k| \leq |\eta_{k-1}^N(g_k)| + \text{osc}(f) |\delta_k|,$$

and inserting $\eta_{k-1}(g_k) = 0$ and $\eta_{k-1}(r_k) = 1$ into the two absolute values gives (57).

For the oscillations, $\eta_k(f) \in [\inf f, \sup f]$ gives $\|f - \eta_k(f)\|_\infty \leq \text{osc}(f)$ and therefore $\|g_k\|_\infty \leq \|r_k\|_\infty \text{osc}(f)$, so $\text{osc}(g_k) \leq 2\|g_k\|_\infty \leq 2\|r_k\|_\infty \text{osc}(f)$. $\square$

We are now ready to prove Theorem 3.

Proof. Define ε_k by the worst-case L^2 error over observables of unit oscillation,

$$\varepsilon_k := \sup\{\|\eta_k^N(h) - \eta_k(h)\|_2 : h \text{ bounded measurable, } \text{osc}(h) \leq 1\}. \quad (58)$$

Then $\varepsilon_k \leq 1$, since $\eta_k^N(h)$ and $\eta_k(h)$ both lie in $[\inf h, \sup h]$. The definition of ε_k implies

$$\|\eta_k^N(h) - \eta_k(h)\|_2 \leq \text{osc}(h) \varepsilon_k \quad \text{for every bounded } h. \quad (59)$$

The proof rests on the recurrence

$$\varepsilon_k \leq \frac{1}{2\sqrt{N}} + 3\|r_k\|_\infty \varepsilon_{k-1}, \quad \text{with the second term absent at } k = 0, \quad (60)$$

which we establish for every $k \geq 0$ and then unroll; Theorem 3 is a consequence of (60).

At $k = 0$, (60) reads $\varepsilon_0 \leq 1/(2\sqrt{N})$. Here η_0^N is the empirical measure of N independent draws from η_0 ,

$$\mathbb{E}[(\eta_0^N(h) - \eta_0(h))^2] = \frac{\text{Var}_{\eta_0}(h)}{N}.$$

Then Popoviciu's inequality on variances (Popoviciu, 1935) bounds the variance $\text{Var}_{\eta_0}(h)$ by $\frac{1}{4} \text{osc}(h)^2 \leq \frac{1}{4}$.

For $k \geq 1$, write A_k for the weighted average (56), so that the identity

$$\eta_k^N(f) - \eta_k(f) = [\eta_k^N(f) - A_k] + [A_k - \eta_k(f)] \quad (61)$$

is exact; the first bracket of (61) is the error injected by the resampling at stage k and the second the error inherited from before it. Recall from Section B.1 that $Z_j^{1:N}$ are the N particles the reduced procedure holds at stage j , and that η_j^N is their empirical measure. Conditionally on $Z_{k-1}^{1:N}$, the stage- k particles $Z_k^{1:N}$ are N independent draws from the weighted distribution $(Z_{k-1}^i, R_k^i)_{i=1}^N$, whose f -average is A_k by (56). Hence $\mathbb{E}[\eta_k^N(f) \mid Z_{k-1}^{1:N}] = A_k$, and again by Popoviciu's inequality on variances,

$$\mathbb{E}\left[(\eta_k^N(f) - A_k)^2 \mid Z_{k-1}^{1:N}\right] \leq \frac{\text{osc}(f)^2}{4N}.$$

This bound is uniform in $Z_{k-1}^{1:N}$, so the tower property gives

$$\|\eta_k^N(f) - A_k\|_2 \leq \frac{\text{osc}(f)}{2\sqrt{N}}. \quad (62)$$

For the second bracket of (61), we employ Lemma 2 to obtain the L^2 error bound

$$\|A_k - \eta_k(f)\|_2 \leq \|\eta_{k-1}^N(g_k) - \eta_{k-1}(g_k)\|_2 + \text{osc}(f) \|\eta_{k-1}^N(r_k) - \eta_{k-1}(r_k)\|_2,$$

where we recall $g_k = r_k(f - \eta_k(f))$, and where the two norms come from taking $\|\cdot\|_2$ in (57) and applying the triangle inequality. Neither g_k nor r_k depends on the particles, so (59) applies to each and gives

$$\|A_k - \eta_k(f)\|_2 \leq [\text{osc}(g_k) + \text{osc}(f) \text{osc}(r_k)] \varepsilon_{k-1}.$$

Finally $\text{osc}(g_k) \leq 2\|r_k\|_\infty \text{osc}(f)$ by Lemma 2, and $\text{osc}(r_k) \leq \|r_k\|_\infty$ because $r_k \geq 0$, so the two oscillations contribute $2\|r_k\|_\infty \text{osc}(f)$ and $\|r_k\|_\infty \text{osc}(f)$ respectively, and

$$\|A_k - \eta_k(f)\|_2 \leq 3\|r_k\|_\infty \text{osc}(f) \varepsilon_{k-1}. \quad (63)$$

Adding (62) and (63) by Minkowski's inequality and taking the supremum over $\text{osc}(f) \leq 1$ yields (60).

It remains to unroll the recurrence. Write $a := 1/(2\sqrt{N})$ and $b_j := 3\|r_j\|_\infty$, and induct on k to prove $\varepsilon_k \leq a \sum_{i=0}^k \prod_{j=i+1}^k b_j$, the empty product being one. At $k = 0$ the right-hand side is a , which is (60) at $k = 0$. Assuming it at $k - 1$, (60) gives

$$\varepsilon_k \leq a + b_k a \sum_{i=0}^{k-1} \prod_{j=i+1}^{k-1} b_j = a \left[1 + \sum_{i=0}^{k-1} \prod_{j=i+1}^k b_j \right] = a \sum_{i=0}^k \prod_{j=i+1}^k b_j,$$

the last equality because the $i = k$ term is the empty product. Taking $k = K$ and applying (59) with $h = f$,

$$\|\eta_K^N(f) - \eta_K(f)\|_2 \leq \text{osc}(f) \varepsilon_K \leq \frac{\text{osc}(f)}{2\sqrt{N}} \sum_{k=0}^K \prod_{j=k+1}^K 3\|r_j\|_\infty.$$

Squaring gives (55). □

B.3 The role of the propagation factor F_Σ

The experiments report $\hat{F}_\Sigma$, the sample counterpart of F_Σ , and the direction of the ordering $F_\Sigma \leq F_{\Sigma+}$ established in Section 4.4 is the first thing to be clear about: F_Σ is a lower bound on the factor of Theorem 2, so substituting it into (23) gives a statement the theorem does not support. F_Σ is not an error bound.

The gap can moreover be made arbitrarily large, because an ESS is a second moment and a supremum is not. Reading the ESS of (25) in measure form, let ν put masses $1 - p$ and p on two points, and let w take the values $(1 - \Lambda p)/(1 - p)$ and Λ on them, with $1 < \Lambda < p^{-1}$, so that $\int w d\nu = 1$ and $w > 0$. Then $\|w\|_\infty = \Lambda$, whereas

$$\frac{1}{\text{ESS}(w)} = \int_{\mathbb{R}^d} w^2 d\nu = \frac{(1 - \Lambda p)^2}{1 - p} + p\Lambda^2. \quad (64)$$

Choosing $p = \Lambda^{-3}$ gives $\text{ESS}(w) \rightarrow 1$ as $\Lambda \rightarrow \infty$ while $\|w\|_\infty = \Lambda \rightarrow \infty$, so a schedule of stages behaving this way would send F_Σ to $K + 1$, the least value it can take, while $F_{\Sigma+} \geq (3\Lambda)^K$ diverges without limit. An ESS weights a large weight by the probability of the set carrying it, and a weight of size Λ on a set of proposal probability Λ^{-3} barely moves the ESS; a supremum is fixed by that weight alone.

For a singular potential the supremum need not be finite. A stage weight is an exponential of a difference of potentials, and the soft cap of a molecular potential slows the growth of the interaction core without bounding that difference, so $\|w_k\|_\infty$ is not controlled by the regularization and may be infinite. Should it be infinite at some stage, $F_{\Sigma+} = \infty$ and the bound of Theorem 2 is unavailable, while a square-integrable weight still has $\text{ESS}(w_k) > 0$ and leaves F_Σ finite. The two factors are governed by different norms of the same weight, the L^2 norm and the L^∞ norm, and the second is the fragile one.

The size of the constant is a third limitation, and the recursion of Section B.2 is where it comes from: each application of (60) re-enters ε_{k-1} , a supremum over all observables of unit oscillation, and the same worst case is taken again at the next stage. Taking $\mu_k \equiv \mu$ and $G_k = \text{id}$ makes every stage weight one, so the scheme is independent sampling followed by K rounds of uniform resampling, and yet $F_{\Sigma+} = (3^{K+1} - 1)/2$ and the constant of (23) grows like 9^K , so the bound says nothing until N reaches that order, while $\text{osc}(\varphi)$ bounds the error at every N . Theorem 2 should therefore be read as establishing the rate $N^{-1/2}$ and the finiteness of the constant, not its size.

F_Σ is nevertheless the reasonable quantity to report. The ESS is the natural second-moment stand-in for the supremum, and the ordering of Section 4.4 is exactly the statement that the L^2 measurement never exceeds the L^∞ one. It is also the classical measurement for importance sampling, since $\int w_k d\nu_k = 1$ gives

$$\frac{1}{\text{ESS}(w_k)} = \int_{\mathbb{R}^d} w_k^2 d\nu_k = 1 + \chi^2(\mu_k \| \nu_k), \quad (65)$$

so F_Σ is assembled from the per-stage chi-squared divergences, and a stage contributes nothing beyond the floor exactly when its proposal equals its target. What it keeps from Theorem 2 is the structure of the constant rather than its value: the same sum over the stage at which an error enters, the same product over the stages it must still traverse, with the problem-independent factors 3 removed.

It is also the more stable of the two to report. Neither factor can be computed directly, and (27) gives a sample counterpart of each, read off the stage weight vectors $\mathbf{W}_j = (W_j^1, \dots, W_j^N)$ that a run already evaluates. The two counterparts do not behave alike: $\hat{F}_{\Sigma+}$ is set by the single largest weight of each vector, while $\hat{F}_\Sigma$ is a second moment of the whole vector. The reported $\hat{F}_\Sigma$ of (27) decreases when a stage ESS improves and increases when the schedule lengthens, combining the two things that make one accepted schedule harder than another.

Section 4.4 already limits its reading twice, in that $\hat{F}_\Sigma$ is not a bound on F_Σ at finite N and that the factor does not measure the difficulty of the target. One consequence for reporting follows. Two runs that traverse the same stage schedule are the most directly comparable, since they share the same least value $K + 1$ and their per-stage ESS values are matched one to one. Across schedules of different length the factor still orders runs by accumulated stagewise weight degeneracy, but the comparison also carries the difference in stage count, and a ratio of two factors is in neither case a prediction of the ratio of two errors.